Instant Semantic Proxy Reconstruction for UAV Digital Twins under Degraded Sensing (SPPA-MVFit)

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Abstract

Remote piloting through a geospatial digital twin is fragile exactly when the twin is stale and the image link is weak: the streamed layer shows the world as captured, and the critical delta—a new tower, vehicles, people near the flight path—is what the static layer cannot contain. We study instant per-object semantic proxy reconstruction for that delta: one detection compiles into a role-labeled eight-part primitive actor (Semantic Primitive Proxy Assembly, SPPA-MVFit) in 9.4 ms median CPU, is emitted as a 1.45 kB update descriptor, and updates the twin in place at telemetry bandwidth as the observation degrades. In preregistered evidence on 240 held-out synthetic actors, the family-conditioned graph improves clean voxel IoU over an equal-budget generic graph by 0.190 (95% CI $[0.181, 0.199]$), and the margin holds under every prespecified observation corruption. Boundary arms map where the prior stops: a wrong family token is worse than none, adversarial instances that violate the structural prior destroy one third of the advantage, and the margin does not replicate on external real meshes (n=52). A case study on a recorded real detector stream (1,902 detections, 2–15 Hz) exercises the same operating point end to end at 11.8 ms median per detection, and detector-token confidence plus prior fit flag wrong tokens (AUC 0.847 $[0.792, 0.898]$ on 217 matched cases). Two exploratory simulated-twin studies add cross-view reconstruction fidelity and camera-less LiDAR-class sensing. Operation is camera-validated: LiDAR and night/fog/smoke are evaluated only through synthetic corruptions and simulated demonstrations; position is assumed given.

Keywords: semantic 3D proxy; multiview silhouette fitting; UAV digital twin; family-conditioned part graph; primitive actor; occupancy IoU; geovisualization of dynamic objects.

1 Introduction

The motivating mission is safe remote piloting through a geospatial twin: a pilot or operator supervises a UAV from a ground station that renders a live Unreal/Cesium 3D Tiles twin—potentially as an immersive VR interface—and must trust what the twin shows. That trust breaks in a specific way: the streamed Cesium layer is frozen at capture time, so it shows the field as it was, not as it is now, and the operationally critical delta—a new or relocated tower, vehicles, animals, people near the flight path—is exactly what the static layer cannot contain. When the live image link is degraded or insufficient to judge the scene, the twin should absorb whatever observation evidence does arrive and reconstruct that delta as explicit per-object volumes, immediately and at telemetry bandwidth. Figure 1 illustrates this twin-delta mission.

Figure 1: The twin-delta mission (conceptual illustration composed from existing project imagery; no new renders or measurements). (a) The real world now contains an intruder tower (Blender photorealistic render). (b) The streamed Cesium/Unreal twin is frozen at capture time and does not contain it. (c) The twin after one detection is compiled into an SPPA proxy and inserted in place; the chip reports the sealed benchmark values for the compact update descriptor (1.45 kB) and the median CPU fit (9.4 ms per object). The imagery is illustrative; descriptor size and fitting time are the sealed values reported in Section 4.

Urban and geospatial digital twins increasingly maintain live object layers—vehicles, field equipment, temporary structures—alongside static city geometry [2, 29]. Each of those layers needs a per-object 3D representation that is cheap to update in place, explicit about uncertainty, and legible at operational zoom levels rather than photoreal. Semantic-telemetry UAV twins make this problem concrete: a UAV-side perception stack reports object tracks, classes, confidence, positions, and approximate geometry, and the ground station must render those objects in an Unreal/Cesium 3D Tiles twin. That architectural choice creates a rendering problem that is separate from the flight system itself: object-level telemetry is not a mesh.

Curated 3D assets are useful when the class is known, the asset library is complete, and runtime loading is acceptable. Those assumptions are weak for field UAV operation. Long-tail labels, ambiguous detections, partial views, occlusion, motion blur, and changing object proportions make a fixed asset library brittle. Neural image-to-3D and text-to-3D methods address a different objective: visual asset generation. They can produce richer geometry, but they depend on usable crops or prompts, GPU inference, and mesh outputs that may be too dense for a portable flight laptop that is also running Unreal, telemetry, video, and tracking under a shared rendering budget.

The Semantic Primitive Proxy Assembly (SPPA) takes a lower-fidelity representation point. Instead of reconstructing the real mesh, it compiles a detection into a semantic part graph made from primitive boxes, cylinders, ellipsoids, cones, and tori. A vehicle becomes a cab, body/cargo region, wheels, and optional top details. A quadruped becomes a torso, head, and legs. A tree becomes a trunk and canopy volumes. Unknown or unsupported labels are not hallucinated into specific objects; they fall back to a generic uncertainty-aware display volume with explicit uncertainty metadata.

The question studied here is deliberately narrow and measurable. Position is assumed given—localization is a separate problem, outside this paper's scope—and what the mission requires is reconstruction fidelity under degraded sensing: does a frozen semantic-family part graph improve lightweight 3D occupancy relative to an input-matched, equal-budget nonsemantic graph—identical silhouettes, optimizer, parameter count, and candidate budget; only the graph changes—when both consume the same calibrated top and side silhouettes, and does the advantage survive the corrupted observations that a weak image link implies? A role-labeled primitive proxy turns detection-level semantics into displayable, analyzable 3D geometry with bounded cost; this paper isolates the occupancy value of the semantic part structure behind that representation under controlled synthetic conditions and maps its behavior as the observation degrades.

The contribution is **family-conditioned SPPA-MVFit**: an instant detection-to-volume path whose semantic prior is validated by a preregistered, equal-budget comparison showing that the family graph improves clean voxel IoU over a generic eight-part graph by 0.190 (95% CI $[0.181, 0.199]$) on 240 held-out synthetic actors, together with boundary analyses that map where the prior helps, where it hurts, and where it stops transferring. The optimizer is deliberately a shared five-parameter global alignment and is not claimed as a novel fitting algorithm; runtime persistence and updateability are reported as separate engineering properties rather than collapsed into a utility score. Concretely, the paper contributes: (i) an instant detection-to-volume path—a reviewed family recipe compiles one detection into a role-labeled eight-part actor in 9.4 ms median CPU per object, emitted as a compact 1.45 kB update descriptor, and sustains 11.8 ms median per detection on a real 2–15 Hz detector stream; (ii) preregistered evidence of volumetric fidelity and of robustness to a weak signal—the +0.190 advantage ($[0.181, 0.199]$) is maintained under all prespecified mask and morphology corruptions, and the frontier where it stops is measured (wrong-family tokens, adversarial family instances, external real meshes, and a real detector stream); and (iii) the twin-delta update contract—role labels, bounded in-place updates, measured 25.8–37.4 kB/s JSON update traffic against a modeled $\geq$250 kB/s video link, and an operational part-query value (F1 0.434)—with its declared limits.

2 Related Work

Text-to-3D and image-to-3D generators target visual asset fidelity from prompts or images [31, 21, 27, 37, 19, 42, 41, 45]. SPPA targets a different runtime point: deterministic actor generation from sparse semantic telemetry, low polygon counts, and cheap per-frame updates.

Monocular and open-vocabulary 3D detection estimate 3D boxes or object state from images [5, 43]; YOLOE supplies approximate open-prompt detection/segmentation evidence rather than universal recognition [39]. Animal pose and reconstruction methods estimate detailed skeletons or category-specific 3D shape [44, 20, 13]. These methods can provide stronger evidence to SPPA, but SPPA itself is a rendering/backend layer, not a detector or pose estimator.

Category-conditioned parametric templates are the closest established paradigm. 3D morphable models fit a low-dimensional learned face prior to image evidence [4], and articulated body and animal models such as SMPL and SMAL fit a category-conditioned parametric template to scans or images [24, 46]. SPPA-MVFit instantiates the same pattern—a family-conditioned template aligned to observation evidence—with an eight-part primitive graph and a deliberately shared five-parameter global alignment. The operating point differs: those models target surface recovery of a specific category, whereas SPPA targets millisecond, track-updateable display actors from sparse UAV telemetry, where the contribution is the bounded runtime contract, the role labels, and the update semantics rather than surface fidelity.

Primitive representations have a long history, from volumetric primitives, recognition-by-components, and superquadrics to learned part decomposition and neural primitive/CSG parsing [3, 36, 30, 38, 25, 35, 10]. Procedural modeling, shape grammars, and learned shape programs generate or abstract structured objects from compact programs [26, 15, 16, 17, 33].

SPPA uses these ideas operationally, but the claimed novelty is the bounded, track-updateable semantic telemetry contract inside an Unreal UAV digital twin.

Recent scene-level systems narrow the gap from other directions: SAM 3D decomposes a single image into composable per-object geometry [34], PartCrafter generates part-structured meshes from one image [22], SuperQuadricOcc reaches real-time semantic occupancy in driving scenes [11], and lightweight urban twins render proxy building geometry at interactive rates [23]; we are not aware of any that constructs role-labeled, metric-scaled object actors from detector tags and sparse silhouettes at millisecond cost. Test-time spatial control steers generative 3D models with coarse primitives [9], which complements rather than replaces a deterministic proxy contract.

Graphics systems already provide level-of-detail, impostors, billboards, streaming tiles, and virtualized geometry [6, 7, 8, 29]. 3D and dynamic scene graphs maintain semantic world structure [1, 32, 40], and Hydra-style systems build hierarchical scene graphs online for mapping and planning [14]; SPPA deliberately keeps its own lightweight contract instead of becoming a node type in such a substrate—a proxy actor is a display-side object with a bounded descriptor/update payload, not a mapping layer. Occupancy grids, OctoMap, ESDF, and TSDF maps support planning and mapping [12, 28]; SPPA should be read as a display actor that can consume such upstream evidence, not as a mapping substrate.

Table 1 positions SPPA-MVFit against these paradigms on the properties that matter for a telemetry-driven UAV twin. To our knowledge, no current paradigm simultaneously (i) consumes detector-level telemetry rather than pixels or prompts, (ii) emits a role-labeled low-polygon actor rather than a dense mesh or a category surface, (iii) updates that actor in place at millisecond CPU cost, and (iv) exposes a bounded descriptor/update contract to the renderer. Reading across the rows: each paradigm fails at least one of the four properties; SPPA-MVFit targets all four. Neural generators fail (i) and (iii); category templates fail (i) and (iv); learned primitive decomposition fails (i) and (iii); scene graphs and occupancy maps solve a mapping problem, not the display-actor problem; and classical level-of-detail techniques require the curated asset that field operation cannot guarantee. SPPA-MVFit is designed for exactly this conjunction.

Table 1: Positioning of SPPA-MVFit against state-of-the-art paradigms for per-object 3D in a telemetry-driven UAV digital twin. *Telemetry input*: consumes detector tags/footprints rather than RGB or prompts. *Role labels*: named semantic parts, not anonymous geometry. *In-place update*: pose/shape updates without regenerating the actor. Cost is the per-object generation cost reported in this paper's measurements or the cited systems' operating points.

Paradigm (representative systems)	Telemetry input	Role labels	In-place update	Per-object cost
Neural image/text-to-3D (TripoSR, Hunyuan3D-2, TripoSG)	no (image/prompt)	no	no	seconds, GPU; 28k–3.4M tris
Category templates (3DMM, SMPL, SMAL)	no (scans/images)	category-specific	partial	offline fitting
Learned primitive decomposition (SuperDec, DualPrim, superfrusta)	no (image/cloud)	implicit parts	no	GPU inference
Scene graphs (Hydra, 3D DSG)	mapping sensors	object level	yes	mapping layer, not a display actor
LOD / impostors / billboards	no (needs assets)	no	yes	negligible, but asset required
Occupancy maps (OctoMap, voxblox)	depth sensors	no	yes	navigation map, not actors
SPPA-MVFit (this work)	**yes**	**yes**	**yes**	**9.4 ms CPU; 8 parts; compact 1.45 kB update descriptor**

3 Materials and Methods

3.1 The SPPA contract and part graph

At time t, SPPA consumes an object state

$$x_t = \{i, \ell, c, b_t, m_t, \xi_t, s_t, \Sigma_t, h_t\},$$

where i is a track identifier, ℓ a detector label, c confidence, b_t a bounding region, m_t an optional mask or footprint, ξ_t a world pose estimate, s_t optional metric dimensions, Σ_t uncertainty, and h_t the track history. The resolver maps ℓ to one of three cases: an `exact_class` label with a fixed archetype; a `keyword_archetype` mapping of an out-of-template label to a finite reviewed archetype, such as vehicle or vertical structure; or a `fallback_unknown_label` rendered as a generic uncertainty-aware volume. The versioned recipe manifest covers 23 archetypes and 95 exact or keyword labels; SPPA can accept an open detector tag at runtime, but the tag is normalized into a reviewed family or a conservative unknown fallback before geometry is emitted. An adversarial resolver audit over 12 compound or misleading labels exercising this routing to the fallback volume is included in RP (`reproducibility/sppa_mvfit/`; *RP* below).

Each archetype a defines a part graph

$$G_a = (V_a, E_a), \quad v_k = (p_k, r_k, \tau_k, \rho_k),$$

where p_k is a primitive type, r_k is its role, τ_k its transform relative to the actor root, and ρ_k role-specific parameters such as dimension priors, color/material role, and confidence tags. The current implementation uses boxes, cylinders, ellipsoids, cones, and tori (Figure 2).

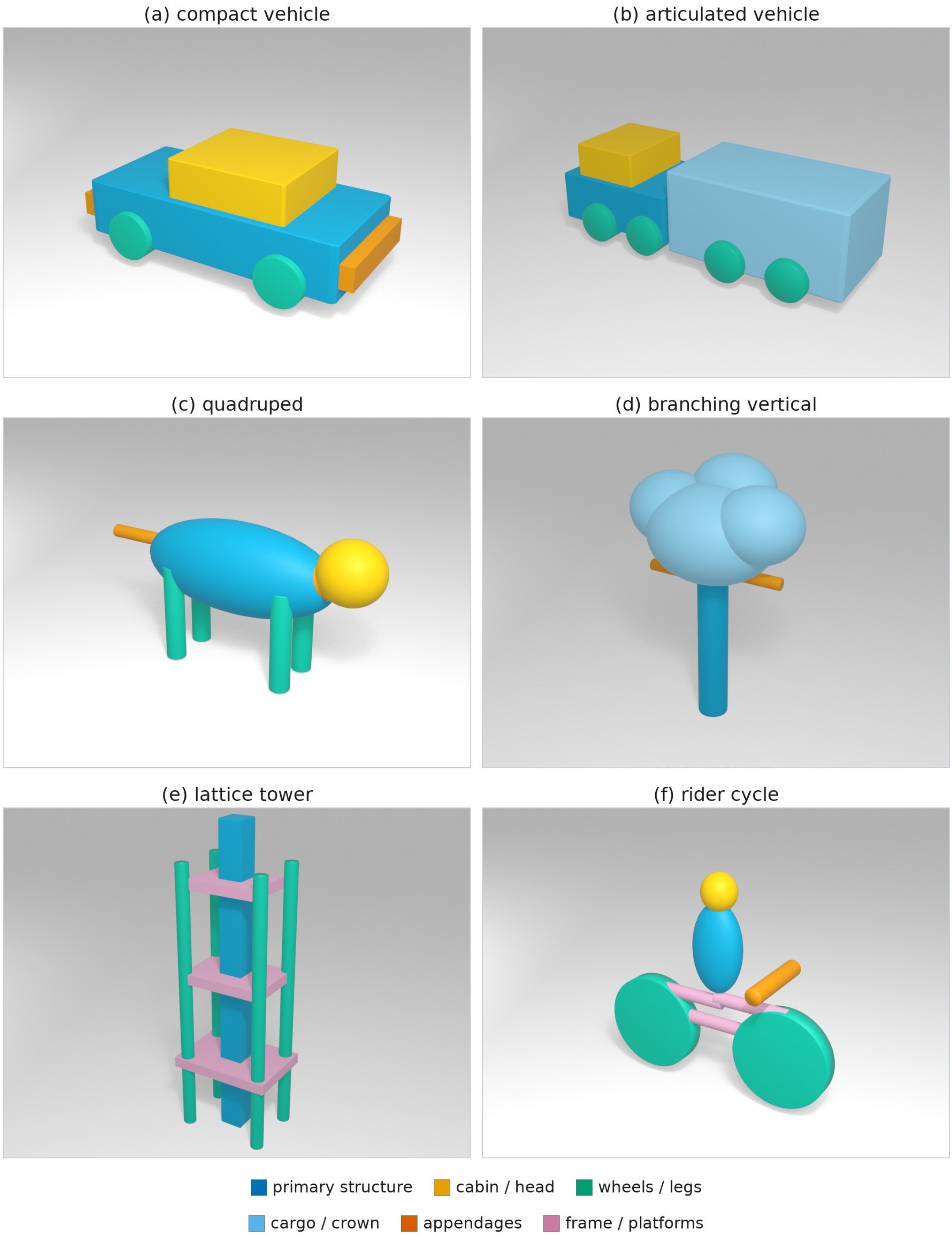

Figure 2: The six benchmark family part graphs used by SPPA-MVFit (`compact_vehicle`, `articulated_vehicle`, `quadruped`, `branching_vertical`, `lattice_tower`, `rider_cycle`): eight primitive slots per family with role-conditioned priors; colors denote semantic roles.

Vehicle adaptation is the clearest implemented case (Figure 3). A long truck and a short truck should not be produced by globally scaling the same box. In the current vehicle grammar, cab and wheel dimensions remain bounded by role priors, while the cargo/body span and axle placement absorb length changes. In the synthetic truck check, two descriptors with equal width/height and different length keep cab and wheel scale fixed at their role priors, while the cargo/body span absorbs the length change. Candidate part decompositions may be drafted offline by a language model, ontology tool, or human author, but only reviewed, versioned, regression-tested recipes are admitted and no language model runs at runtime.

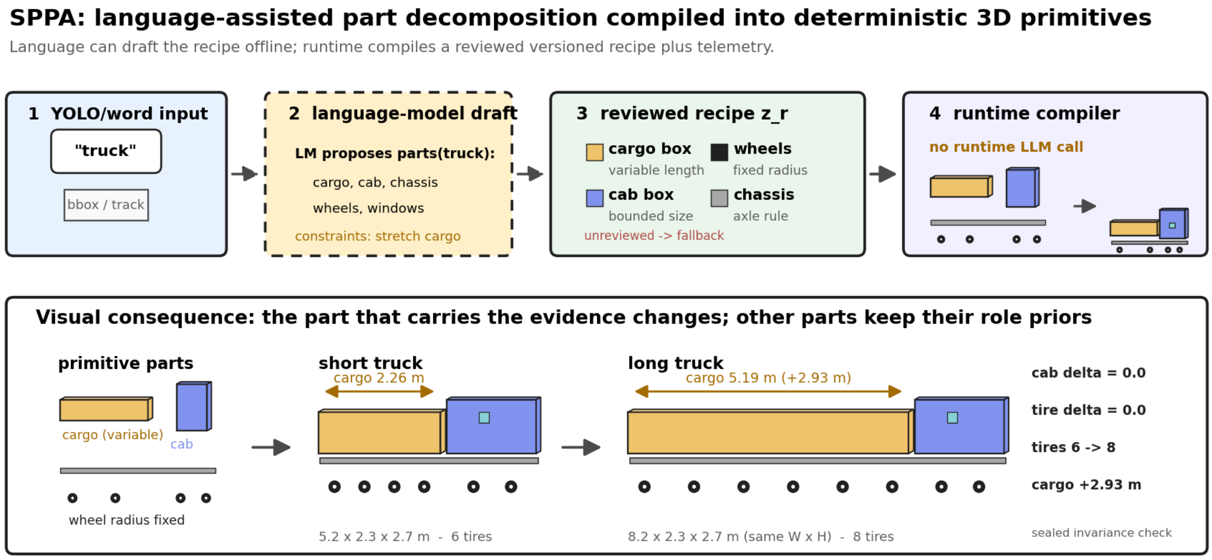

Figure 3: SPPA semantic-label-to-parts-to-3D mechanism for the truck archetype. A language model, ontology tool, or human author may draft candidate part roles offline, but runtime execution uses only a reviewed versioned recipe z_r: an input label or word activates an offline draft, the draft is reduced to reviewed part roles and bounded update rules, and runtime compiles those roles into primitive geometry. The short and long proxy variants illustrate the implemented role-specific rule: cargo length changes while cab and wheel priors remain bounded, so the object is not represented by root-scaling a single box.

Pose, yaw, and updates. The descriptor separates actor pose ξ_t from part parameters q_t. Pose changes are per-frame updates; topology regeneration is avoided unless the class/archetype changes or new evidence crosses a configured threshold. Yaw ambiguity is handled explicitly: a top-down footprint or PCA orientation gives an axial orientation modulo π, and signed yaw requires velocity, asymmetry, multi-frame evidence, or an external heading source; otherwise the descriptor sets `yaw_ambiguous=true`.

3.2 End-to-end pipeline: two input paths, one builder

Both input contracts terminate in the same deterministic builder; they differ only in how much evidence reaches the recipe. The tag/text path compiles a reviewed recipe at its prior dimensions: normalize the label through the reviewed ontology, select the frozen part graph G_a and its prior role parameters, emit role-labeled primitives with connector cylinders for semantically attached parts, and emit the versioned descriptor consumed once per track by the renderer. The detector/image path inserts four evidence stages before the same builder: (1) **Observe**—detector label, confidence, bbox, and native mask polygon are stored in a `SPPA-OBS-0.1` record with declared camera/flight geometry and a covariance-style uncertainty envelope; (2) **Project**—the mask polygon is projected through the declared camera model into an oriented ground footprint, keeping the axis-aligned mask box and a PCA axial orientation as audit evidence; (3) **Scale**—with declared metric calibration the footprint becomes a metric length/width/height proposal; without calibration the mask remains image-space evidence and cannot change metric geometry; (4) **Gate**—a class-aware constraint-fusion step accepts plausible scale, clamps implausible length/width ratios through the family aspect prior, withholds unsafe image-derived pose/yaw from the runtime descriptor, and records every decision with its reason. Because both paths share the builder and the descriptor contract, their outputs are directly comparable.

The evidence contract is sensor-agnostic by design. What the Project and Scale stages consume is an oriented ground footprint geo-projected from whatever sensor produced the observation, together with its declared geometry and uncertainty envelope; any sensor that delivers such a footprint enters the same contract and the same builder. In this paper the camera path is the validated one—it is exercised by the real-image probes of Section 4.9 and by the recorded detector stream of Section 4.10. LiDAR is an interchangeable input by design: a LiDAR cluster reduces to the same geo-projected footprint and enters the same gate. No hardware LiDAR measurement is reported here; the camera-less path is exercised only in the simulated-LiDAR twin demonstration of Section 4.12, and night, fog, or smoke operation is likewise evaluated only through the prespecified synthetic mask corruptions of Section 4.2 and the simulated degraded arm of Section 4.12, not through measured degraded-weather data.

Worked example (tractor-trailer probe). The real crop yields YOLOE evidence `vehicle` at confidence 0.47 with a 54-point native mask. Projection proposes a raw footprint of $16.85 \times 6.21 \times 3.40$ m; the gate applies soft low-confidence fusion, clamps the footprint through the vehicle aspect prior to $12.11 \times 3.76 \times 3.40$ m, withholds image-derived pose, and refines the weak label only to the conservative `articulated_vehicle` family. The shared builder then emits a 1,904-triangle proxy (evidence in RP).

The full chain is written out step by step in RP, and Figure 4 traces one measured detection of the real stream of Section 4.10 through every stage of that chain.

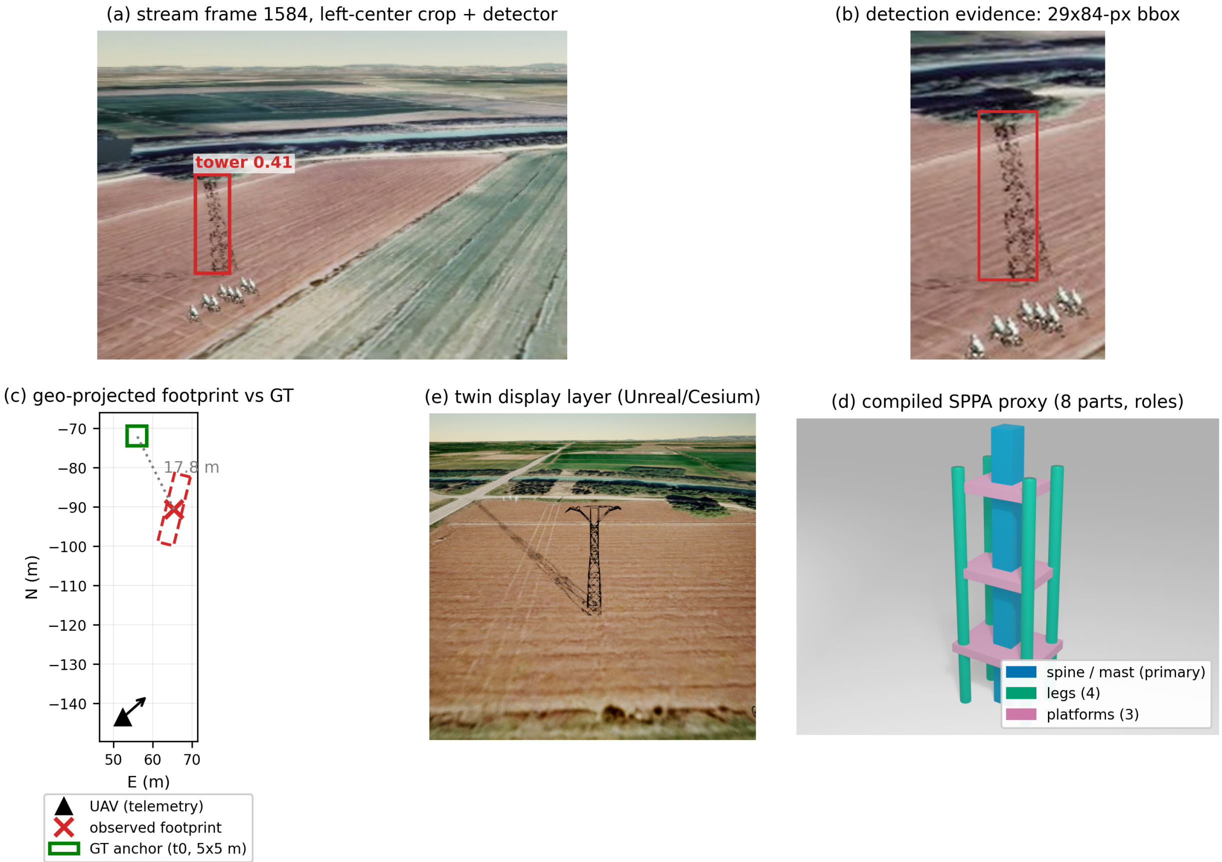

Figure 4: Worked end-to-end example on measured field data (frame 1584 of the real stream, Section 4.10). (a) The detector reports a `tower` at confidence 0.41, the only hypothesis in this frame. (b) Detection evidence: the 29×84-px detector bbox, cropped and magnified. (c) The geo-projected footprint (18.0 × 4.2 m observed) against the exact simulator anchor of tower `t0`; the localization gap is set by the monocular observation geometry and is quantified for the whole stream in RP. (d) The compiled eight-part `lattice_tower` proxy with semantic roles (spine, legs, platforms), placed directly below the real crop for silhouette comparison. (e) The Unreal/Cesium twin layer in which the emitted actor is displayed.

3.3 Descriptor and runtime backend

A `SPPA-DESC-0.2` record carries the resolved family, the role-labeled part list, and the evidence, uncertainty, and material tags needed to build the actor once per track; a `SPPA-UPD-0.2` record separates create/regenerate payloads from the pose/no-op/shape updates applied in place (the field-level record contents and the `descriptor_id` topology-cache identity rule are detailed in RP). The design goal is to generate once per track/archetype, update pose/material/uncertainty every frame, and apply shape-parameter changes in place when the part topology is unchanged. An optional Unreal Engine 5.7 backend ingests the same descriptors alongside curated assets, which remain the default; the SPPA backend constructs primitive components from the `parts` array, applies update packets, and rejects invalid JSON or mismatched descriptor identifiers without changing the current actor.

3.4 Confirmatory protocol

Every compared method receives the same case identifier, morphology-family token, metric world box, and two 96 × 96 orthographic binary silhouettes (top and side). No method receives RGB pixels, detector scores, source parameters, ground-truth occupancy, evaluation projections, or real or replay telemetry. An actor is an ordered list of eight primitive slots (boxes, cylinders, ellipsoids).

Family graphs and a single `generic` graph are stored as static JSON. The only builder used by text-only SPPA, Generic-MVFit, and SPPA-MVFit is

$$A = \mathrm{Build}(g, \theta), \qquad \theta = (\log s_x, \log s_y, \log s_z, s_{\mathrm{sec}}, o_{\mathrm{sec}}), \tag{1}$$

with identical bounds for every graph g. Text-only SPPA uses the family graph at the default θ_0; Generic-MVFit and SPPA-MVFit optimize the same five parameters under the same objective and candidate budget, and only the graph identity differs (`generic` vs family). The fitting loop is written out exactly in Figure 5. Baselines include axis-aligned box, ellipsoid, capsule, billboard, and nonsemantic visual hull from the same masks.

Algorithm 1 — SPPA-MVFit: 31-candidate silhouette fit (identical for every compared graph g).

Input: graph g; binary silhouettes S_{top}, S_{side} (96×96); metric world box.
Parameters: $\theta = (\log s_x, \log s_y, \log s_z, s_{\mathrm{sec}}, o_{\mathrm{sec}})$, shared by all eight slots.
Objective: $J(\theta) = 0.5\,(1 - \mathrm{IoU}_{\mathrm{top}}) + 0.5\,(1 - \mathrm{IoU}_{\mathrm{side}}) + \varepsilon$·centering, evaluated on the actor's projections against the observed silhouettes.

1. Initialize θ_0 from the observed mask extents; evaluate $J(\theta_0)$ (1 evaluation)
2. For each step fraction $f \in \{0.10, 0.03, 0.01\}$ (three shrinking sweeps): for each of the five parameters, evaluate J at the current best θ with that parameter moved by $\pm f$ times its allowed range, and keep the better value ($3 \times 5 \times 2 = 30$ evaluations)
3. Return θ^*. Total cost: exactly $1 + 30 = 31$ objective evaluations for every method—the budget is part of the controlled comparison, not a tuned choice.

Figure 5: The complete fitting loop used by SPPA-MVFit and Generic-MVFit. The only difference between the two methods is the graph g passed to the shared builder; the parameterization, bounds, objective, initialization, and candidate budget are identical.

The fitter is deliberately a five-parameter global alignment shared by all slots; the contribution is the family-conditioned representation and the runtime contract, not the optimizer (a fitted example sequence is included in RP).

The `generic` graph was designed by the authors as a plausible family-agnostic prior: eight symmetric ellipsoid slots with no family-specific topology, proportions, or role labels, frozen before the benchmark was run. It is hand-crafted, not learned, not random, and not the mean of the six family graphs. Because the same team authored the family graphs and this generic competitor, three alternative generic graphs were registered in writing before any measurement and are evaluated in Section 4.6; the shared-authorship threat is discussed in Section 5.

The primary hypothesis compares clean-mask 64-cubed voxel IoU of SPPA-MVFit against Generic-MVFit. Analysis units are 240 source actors (20 per family in each of two strata). The confirmatory estimand is the equal-stratum mean paired difference. H1 passes only if the lower endpoint of a two-sided 95% stratified actor-bootstrap percentile interval (10,000 resamples, seed 77157) exceeds +0.030. The margin and the power analysis behind the sample size were fixed before evaluation: with paired-difference SD 0.12, n=240 actors give approximately 90% normal-approximation power to distinguish a true mean of +0.055 from the +0.030 margin. The full protocol was preregistered and sealed before evaluation; the seed provenance, amendment log, and freeze records are given in RP. Two amendments to the sealed protocol are disclosed: Amendment 03 substituted the planned external review of the protocol (not obtainable) by an internal triple-role self-audit, and Amendment 04 re-derived the secondary inference after the seal (null-centered bootstrap with Holm adjustment). Development results were used only to implement the frozen package and are not confirmatory.

Source actors come from two strata: CSG-ID unions of solids with family topology, and implicit-OOD occupancy from superelliptic extrusions, taper/twist, toroidal sections, and spline-distance tubes. Source and method modules do not import each other. The held-out set is developer-held-out synthetic geometry, not an externally blinded public UAV benchmark.

All confirmatory and post-hoc fits ran on one benchmark workstation (AMD Zen 5, 32 logical CPUs, Windows 11, CPython 3.12.6); the external neural-generator runs of Section 4.8 used an RTX 5090 32 GB GPU on the same machine.

3.5 Evaluation data

The synthetic benchmark covers six families selected before freezing for morphological coverage of the ground-object envelope a UAV twin must display: compact rigid bodies (`compact_vehicle`), elongated articulated bodies (`articulated_vehicle`), legged organic forms (`quadruped`), branching vertical growth (`branching_vertical`), thin lattice structures (`lattice_tower`), and rider-plus-cycle combinations (`rider_cycle`), with 20 actors per family in each of the two strata (n=240). A post-hoc exploratory drop-one-family re-analysis (Section 4.3) confirms the headline difference is not carried by any single family: excluding any one family leaves the paired difference between 0.152 and 0.214, always above the +0.030 margin. As an external case study outside the design distribution, the same methods were run on n=52 real third-party meshes (Objaverse LVIS subset and ModelNet40) under a class-to-family mapping frozen in writing before any evaluation (Section 4.7). Real-image probes (a road cyclist, a mountain electrical tower, a mountain tractor, a tractor with trailer, and two real flight photos of a tower and a tractor recorded for this paper on 2026-07-21) exercise the dual-input contract of Section 3.2 with a YOLOE-26s open-vocabulary detector (checkpoint `yoloe-26s-seg.pt`) as approximate semantic evidence (Section 4.9).

4 Results

4.1 Primary confirmatory result

Table 2 reports the confirmatory decision: H1 passes, with mean ΔIoU = 0.190 and 95% CI $[0.181, 0.199]$. The CSG-ID stratum mean difference is 0.209 and the implicit-OOD stratum mean difference is 0.172; neither is non-positive, so a cross-stratum generalization warning is not triggered by the protocol rule. The OOD stratum is also the in-benchmark guard against a design-decided result: its actors are built by a different generator (superelliptic extrusions, taper/twist, toroidal sections, spline-distance tubes) that does not instantiate the CSG family topology, and the margin persists there. The structural-favor threat itself is bounded by measurement in Sections 4.5 and 4.6 and discussed in Section 5. Figure 6 and Table 3 report the paired difference in every family-by-stratum cell. All twelve point estimates are sealed and positive: the largest gain is `rider_cycle` CSG-ID (+0.458, CI $[0.402, 0.508]$) and the smallest is `compact_vehicle` implicit-OOD (+0.043, CI $[0.028, 0.057]$). The post-hoc exploratory drop-one-family re-analysis of Section 3.5 leaves the paired difference between 0.152 and 0.214, so no single family carries the headline.

Table 2: Preregistered primary endpoint on the held-out synthetic test (n=240 actors). Provenance: `synthetic_geometry`. H1 passes only if the stratified bootstrap 95% lower bound exceeds +0.030.

Quantity	Value
Primary mean Δ IoU (SPPA-MVFit − Generic-MVFit)	0.190
Stratified bootstrap 95% CI	[0.181, 0.199]
Prespecified superiority margin	+0.030
H1 decision (lower bound > margin)	PASS
Actors (analysis units)	240
CSG-ID stratum mean Δ	0.209
Implicit-OOD stratum mean Δ	0.172
Resolution sensitivity 48/64/80	PASS

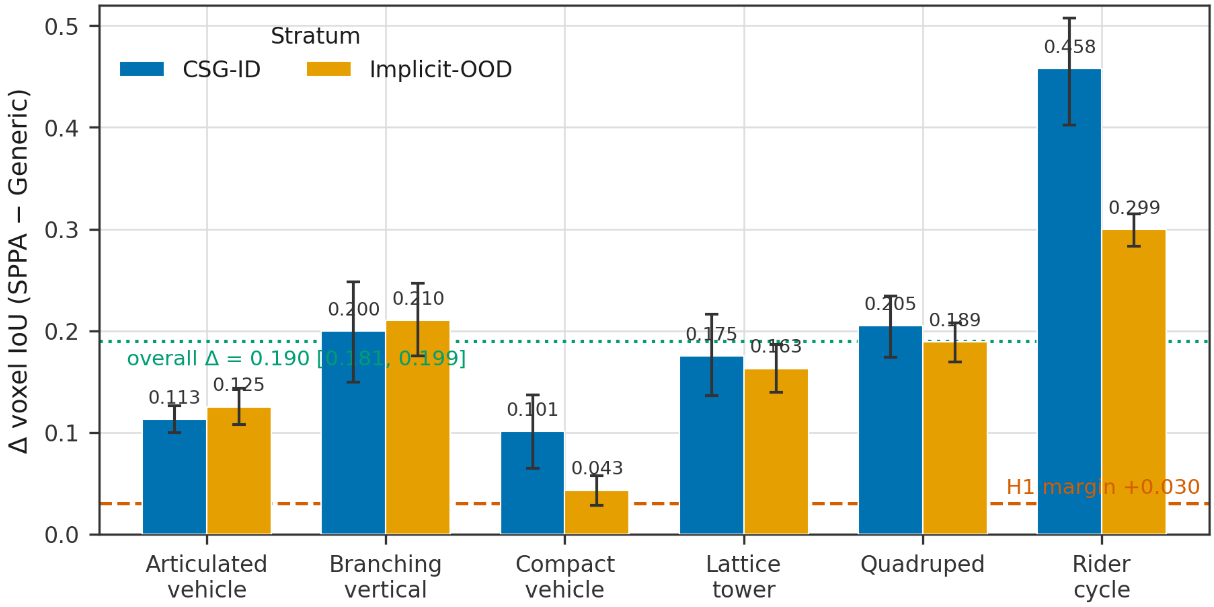

Figure 6: Paired clean-mask mean IoU difference of SPPA-MVFit minus Generic-MVFit by morphology family and source stratum (n=240). All twelve cells are positive; the equal-stratum aggregate is the primary estimate of Table 2.

Table 3: Paired SPPA-MVFit minus Generic-MVFit mean IoU difference by morphology family and source stratum (preregistration §6 obligation; n=20 actors per cell). Point estimates are sealed (`confirmatory_summary.json`); the per-cell 95% CIs are NEW post-hoc within-cell bootstrap intervals (10,000 resamples over the 20 actors of each cell, seed 77157, percentile)—the sealed package stores no cell CIs. The aggregate row is the sealed primary estimate; the drop-one-family range is post-hoc and not sealed.

Family	CSG-ID ΔIoU [95% CI]	implicit-OOD ΔIoU [95% CI]
articulated_vehicle	+0.113 [0.100, 0.127]	+0.125 [0.107, 0.144]
branching_vertical	+0.200 [0.151, 0.247]	+0.210 [0.175, 0.246]
compact_vehicle	+0.101 [0.064, 0.137]	+0.043 [0.028, 0.057]
lattice_tower	+0.175 [0.136, 0.215]	+0.163 [0.140, 0.186]
quadruped	+0.205 [0.174, 0.234]	+0.189 [0.169, 0.208]
rider_cycle	+0.458 [0.402, 0.508]	+0.299 [0.282, 0.315]
Aggregate (sealed)	+0.190 [0.181, 0.199]	(n = 240, sealed stratified bootstrap)
Drop-one-family range (post-hoc)	0.152–0.214	(exploratory artifact t5_drop_one_family)

Table 4 reports clean-condition method means and wall-clock costs, and Table 5 the sealed paired differences of SPPA-MVFit against every baseline under the same stratified bootstrap construction. SPPA-MVFit exceeds SPPA text-only (mean $\Delta = 0.130$, CI $[0.116, 0.144]$; the preregistered `curated_family_template` row is this identical method, deduplicated under Amendment 01) and all lightweight geometric baselines. The margin over the strongest baseline is much smaller than the headline: the sealed paired advantage over the nonsemantic visual hull [18] is +0.036 (CI $[0.027, 0.044]$), roughly one fifth of the +0.190 headline, and visual hull is denser occupancy geometry without a role-labeled lightweight actor contract, so it is not presented as a defeated runtime competitor. The summary statistics invert by construction: the hull's median (0.567) exceeds SPPA-MVFit's (0.563) while its mean (0.522) is lower—a tail asymmetry in which the paired mean advantage is carried by a right tail of actors where the eight-part contract substantially out-fits the hull, not by a uniform per-actor shift. Median single-call descriptive SPPA-MVFit time is 9.4 ms (p95 10.6 ms), below the secondary local-cost gates of 100 ms median / 250 ms p95; under the identical 31-candidate budget the SPPA-MVFit median fit (9.43 ms) is slightly lower than Generic-MVFit's (11.34 ms), a wall-clock difference the package records without attributing it to a single cause. A resolution-sensitivity check at grids 48/64/80 on the prespecified actor subset changes the paired difference by at most 0.0094, below the 0.015 protocol threshold. Secondary intervals are unadjusted per-comparison intervals; confirmatory inference is H1 only (Amendment 04 reports Holm-adjusted p-values, ≈ 0 for all six baseline comparisons, in RP).

Table 4: Clean-mask method summary on the held-out test (n=240). Mean/median are 64-cubed voxel IoU. Timings are single-call wall-clock milliseconds on the benchmark machine and are not Unreal frame times.

Method	Mean IoU	Median IoU	Median ms	p95 ms
SPPA-MVFit	0.557	0.563	9.43	10.60
Generic-MVFit	0.367	0.390	11.34	11.81
SPPA text-only	0.427	0.441	0.77	1.81
Visual hull	0.522	0.567	0.22	0.24
Ellipsoid	0.342	0.292	0.50	0.55
Capsule	0.325	0.260	0.95	1.07
Axis-aligned box	0.248	0.190	0.23	0.27
Billboard	0.173	0.148	0.23	0.26

Table 5: Sealed paired differences of SPPA-MVFit against every baseline on the held-out test (clean 64-cubed voxel IoU, n=240, stratified bootstrap, 10,000 resamples, seed 77157; all deltas sealed in `confirmatory_summary.json`). Means and medians are recomputed read-only from the sealed `raw_metrics.csv`.

Method	mean IoU	median IoU	SPPA−method Δ [95% CI] (sealed)
sppa_mvfit	0.557	0.563	
nonsemantic_visual_hull	0.522	0.567	+0.036 [0.027, 0.044]
sppa_text_only	0.427	0.441	+0.130 [0.116, 0.144] (ablation)
generic_mvfit	0.367	0.390	+0.190 [0.181, 0.199] (primary)
ellipsoid	0.342	0.292	+0.216 [0.207, 0.224]
capsule	0.325	0.260	+0.233 [0.224, 0.241]
bbox	0.248	0.190	+0.310 [0.301, 0.318]
billboard	0.173	0.148	+0.384 [0.375, 0.393]

4.2 Robustness to degraded observations (preregistered conditions)

The mission premise is a weak signal: the observation that reaches the twin may be noisy, partially occluded, or morphologically inconsistent with the family prior. This arm measures exactly that regime. The observation conditions—mask corruption, partial occlusion, and mild and moderate morphology corruption—were prespecified in the protocol; the inference over the sealed data is post-hoc under Amendment 04 (Table 6; Figure 7). The margin is exceeded under all five conditions, and the contrast between corruption types is the headline: mask corruption costs the paired difference only $\approx$0.001 IoU (0.189 against 0.190 clean) and partial occlusion the same (0.189); mild morphology corruption leaves the delta at 0.163, and only the moderate level—evidence that contradicts the graph prior—cuts it to 0.118 (CI $[0.106, 0.131]$), still far above the prespecified +0.030 margin. Missing or noisy evidence is therefore nearly free for this method; contradictory evidence is what degrades it, and even that leaves the prespecified margin intact.

Table 6: Robustness of the primary comparison to degraded observations (post-hoc analysis of the sealed confirmatory data under the preregistered conditions; n=240 per condition). Method means and paired differences under the five prespecified observation conditions. The prespecified +0.030 margin is exceeded under every condition. Per-condition means of all eight methods are in RP.

Condition	SPPA-MVFit	Generic-MVFit	Δ paired	95% CI
Clean	0.557	0.367	0.190	[0.181, 0.199]
Mild morph.	0.512	0.349	0.163	[0.152, 0.173]
Moderate morph.	0.418	0.299	0.118	[0.106, 0.131]
Partial occl.	0.545	0.355	0.189	[0.179, 0.200]
Mask corrupt.	0.555	0.367	0.189	[0.179, 0.197]

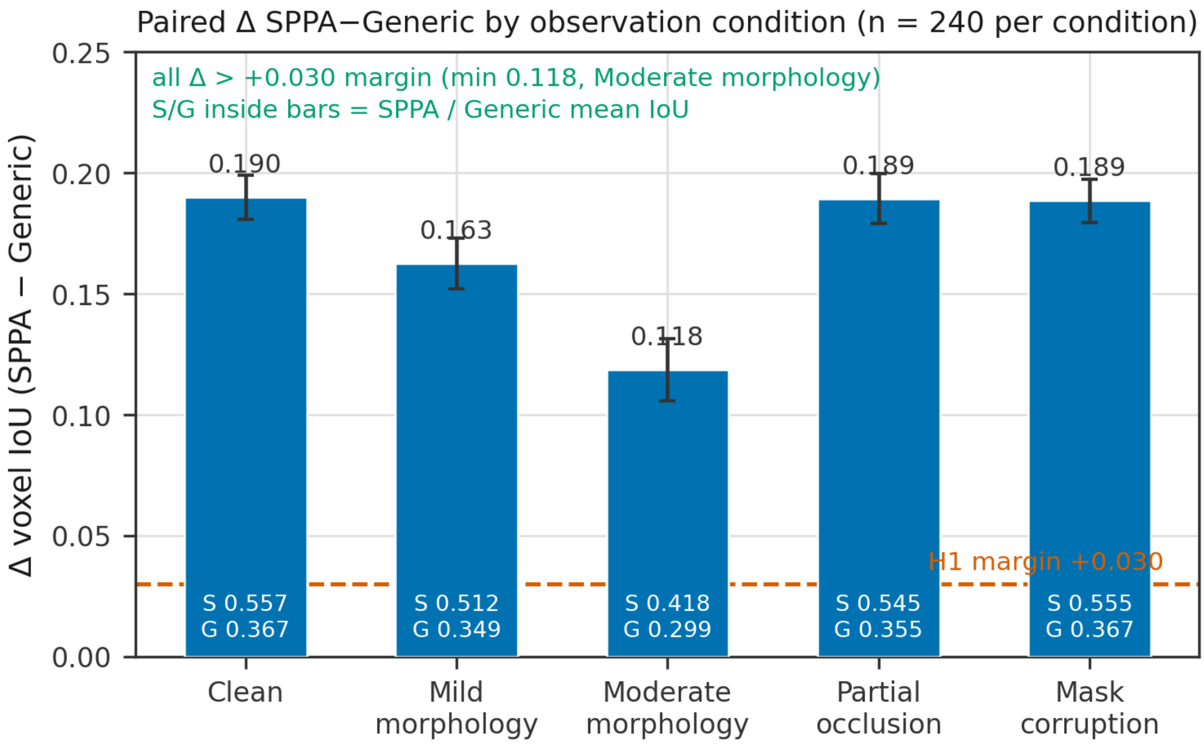

Figure 7: Paired SPPA-MVFit minus Generic-MVFit mean IoU difference with 95% bootstrap intervals under the five prespecified observation conditions (n=240 per condition). The dashed line marks the prespecified +0.030 margin; it is exceeded under every condition.

Exactly 5 of 240 clean cases fall below the prespecified failure threshold (voxel IoU < 0.25); all five are `lattice_tower` CSG-ID cases (worst 0.147 and 0.148). Inspection attributes the failures to sub-voxel geometry: the failing towers have leg and plate thicknesses of 0.09–0.37 world units against a $0.15\times0.10\times0.10$ voxel cell, and the seven actor methods score at most 0.24 on these actors (visual hull at most 0.43)—a resolution effect shared by every method, not a family-prior failure. Convergence telemetry supports the same reading: 88.7% of SPPA-MVFit cases still improve in the final sweep and its θ never sits at the frozen bounds, whereas Generic-MVFit pins θ at a bound in 57.9% of cases—evidence of a mismatched prior demanding out-of-range excursions, not of SPPA overfitting. Surface metrics agree with the volume ranking (Chamfer and F-score tables in RP).

4.3 Representation versus fitting: the graph-by-fitting decomposition

A post-hoc exploratory analysis completes the graph-by-fitting factorial: the previously missing cell, the generic graph at the default θ_0, scores 0.180. The family graph with no fitting at all (0.427) already exceeds the fully fitted generic competitor (0.367). Fitting adds +0.187 on the generic graph and +0.130 (CI $[0.117, 0.143]$) on the family graph, with a negative interaction (-0.058). The H1 headline therefore decomposes into a larger representation effect—graph effect at no fit +0.248, CI $[0.234, 0.261]$—plus a smaller fitting contribution on top of it, and the claim of this paper is accordingly about the family-conditioned representation, not about fitter sophistication.

4.4 Wrong-family tokens

The confirmatory benchmark supplies the correct family token, so it does not measure the principal failure mode of a family-conditioned method. As a post-hoc exploratory arm, each held-out actor was therefore also fitted under each of the five incorrect family tokens (1,200 additional deterministic fits through the same fitting path; Figure 8). A wrong token is far worse than no family prior at all: the wrong-token mean is 0.205 against Generic-MVFit 0.367, a paired difference of -0.162 (CI $[-0.167, -0.158]$), with 98.3% of actors scoring below their own generic fit, and even the per-actor *best* wrong token (0.361) does not beat the generic mean. The deployed contract already routes unsupported labels to a generic fallback volume and refines low-confidence detector labels only to conservative families; this result makes the symmetric design implication explicit—a family token should be accepted only with adequate detector confidence or agreement, and otherwise the fit should degrade to the generic graph rather than commit to a mismatched family.

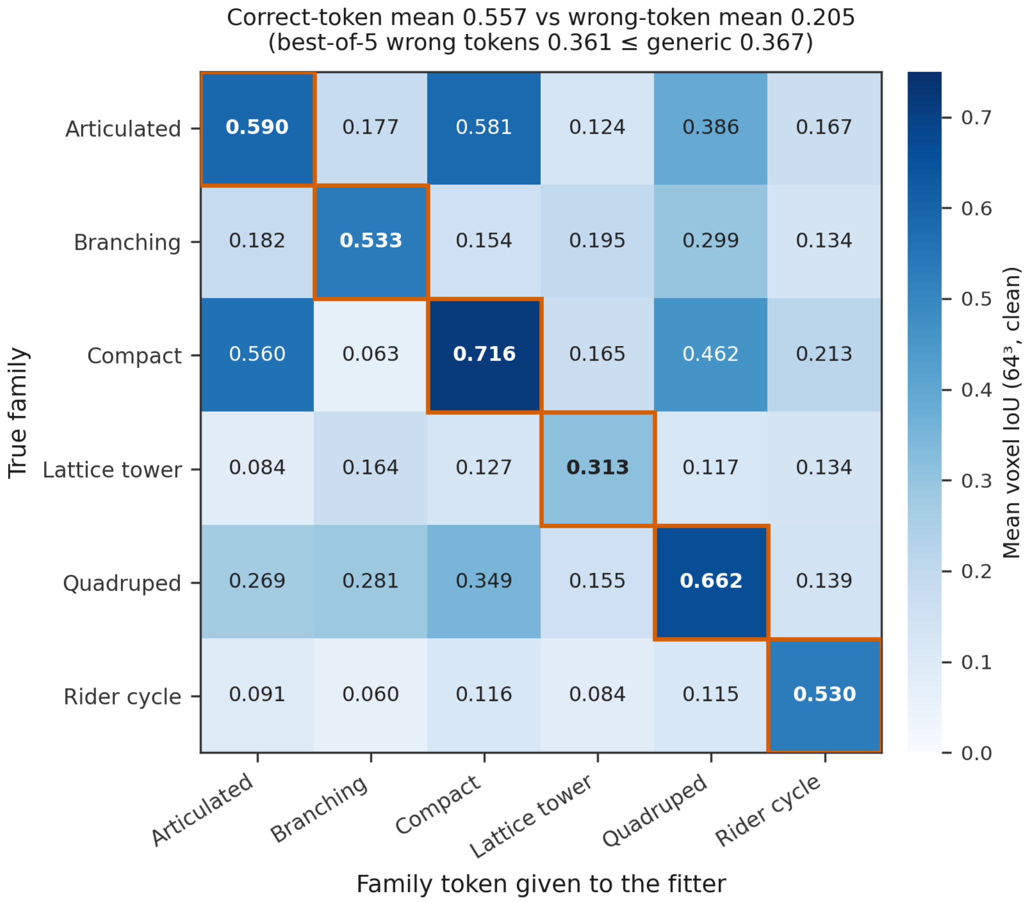

Figure 8: Wrong-family analysis (clean condition, n=240, mean voxel IoU). Rows: true family; columns: family token supplied to the fitter. The diagonal is the correct token. Off-diagonal cells fall below the diagonal and, with few exceptions, also below the generic-graph mean of 0.367; the exceptions cluster on the morphologically close vehicle families (compact/articulated) and on the quadruped token applied to vehicle rows.

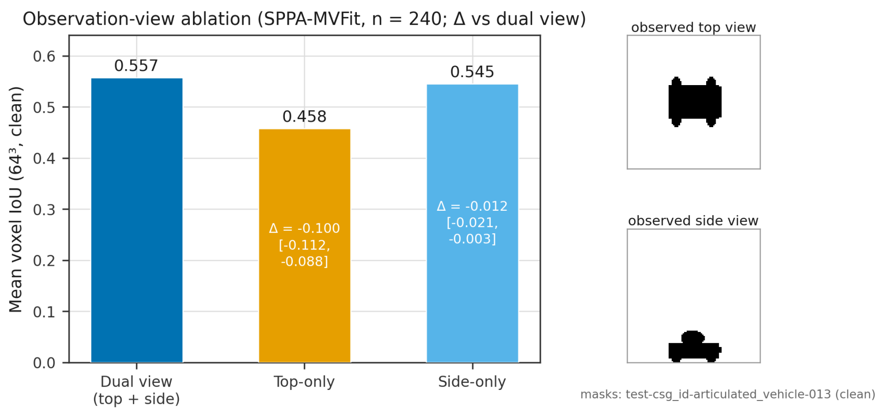

Figure 9: View ablation of SPPA-MVFit (clean condition, n=240, correct family token): dual-view, top-only, and side-only mean voxel IoU with 95% bootstrap intervals.

4.5 Adversarial family instances

A structural concern with the primary comparison is that it is nearly guaranteed: the family graph encodes the same part topology that the CSG-ID generators instantiate, so a positive H1 could be decided by design rather than by measurement. The wrong-family arm already shows the prior can be actively harmful; here we probe the complementary edge. A post-hoc exploratory stratum of 120 adversarial actors (20 per family, twelve documented violation types, design frozen before measurement) keeps each family's semantic class but violates its structural prior: rearward cabs, split cargo beds, six-legged and asymmetric quadrupeds, out-of-order platforms, a 25° leaning lattice tower, sidecar cycles, and cargo over-scaled to 250% of the prior. Under the identical clean-mask protocol, the paired SPPA advantage over Generic-MVFit falls from +0.209 on the same actors unviolated (i.e., the CSG-ID stratum) to +0.141 (CI $[0.125, 0.157]$): violating the prior destroys about one third of the advantage ($\Delta\Delta = -0.068$, CI $[-0.087, -0.048]$) and SPPA-MVFit loses at actor level in 8.3% of cases. Two violation types reach statistical ties with both methods near the floor—the leaning tower (+0.009, CI $[-0.008, +0.026]$; a 25° shear is not expressible by axis-aligned part scaling) and the cascade crown (+0.007, CI $[-0.015, +0.029]$)—while others leave the advantage intact or larger (sidecar +0.320). The family prior is therefore neither a tautological win nor a fragile one: its value degrades measurably and predictably with prior violation, and the failure zones are identifiable (table in RP).

4.6 Ablations and sensitivity analyses

View ablation. The objective consumes a top and a side silhouette. In this post-hoc exploratory ablation, refitting with only the top view costs −0.100 IoU (CI $[-0.112, -0.088]$); refitting with only the side view costs only −0.012 (CI $[-0.021, -0.003]$) (Figure 9). The side view carries nearly all of the 3D information for this object envelope, and even the degraded top-only mode (0.458) stays above dual-view Generic-MVFit (0.367) and the box baselines. A side view is not available from a pure nadir pass; the top-only arm therefore quantifies exactly the nadir observation regime—the footprint-only mode of the operational pipeline (Section 3.2)—for the fitting path.

Oriented bounding box baseline. A rotation-aware oriented bounding box baseline stays within 0.004 IoU of the axis-aligned box on this axis-aligned benchmark, so orientation freedom does not close the representation gap (full analysis in the reproducibility package).

Generic-graph design sensitivity. Because the authors designed the generic graph (Section 3.4), three alternative generic graphs were registered in writing before any measurement and evaluated under the same protocol: G2 (box/cylinder chassis), G3 (vertical ellipsoid stack plus legs), and G4 (slot-wise mean of the six family graphs, a fully mechanical rule). All three score *below* the original generic (table in RP): against these alternatives the measured SPPA advantage is, if anything, conservative.

Optimizer budget. The 31-candidate budget sits at the knee of the budget curve: doubling to 61 yields a non-significant +0.002 (CI $[-0.001, +0.005]$) for twice the compute (full sweep in RP). Sweep timings come from a separate post-hoc re-run on the same machine; IoU values are bit-exact with the sealed run, which reports a 9.4 ms median.

Role-aware IoU. Occupancy IoU does not test the role labels themselves. As a first quantitative check, a slot-to-component mapping against the ground-truth generators was frozen in writing before any computation and restricted ex-ante to the CSG-ID stratum (120 actors, 920 matched slot–component pairs). Matched pairs reach 0.545 coverage and a role IoU of 0.319, against 0.053 for a random-shuffle control (cyclic shuffle 0.017): a paired role-assignment advantage of +0.265 (CI $[0.250, 0.281]$; table in RP). This is descriptive evidence on a frozen mapping over one stratum, not a confirmatory endpoint.

Part-query task. To operationalize what role labels are worth, a part-query task was frozen in writing before measurement on the same stratum: given a target role (*cargo*, *wheels*, *cab*, *legs*, *hitch*, *platforms*), each method must return the voxels of that part. SPPA-MVFit returns its role-labeled slot; controls are Generic-MVFit with its own slot mapping and two frozen visual-hull heuristics (largest connected component; lower half). Over 887 scorable queries (33 of 920 excluded for sub-voxel ground truth, documented with a sensitivity analysis), SPPA-MVFit reaches query F1 0.434 versus 0.145 (generic), 0.111 and 0.087 (hull heuristics), with normalized part-centroid error 0.055 versus 0.177–0.249; the hull *covers* (recall 0.96) but cannot *select* (precision 0.07). Structural part counting is exact in 70% of actors versus 0% for the hull, and the frontier is clear: role value concentrates on small, numerous parts, with residual failure zones where graph arity and ground truth disagree (table in the reproducibility package).

4.7 External sanity check on real meshes

As a post-hoc exploratory case study outside the design distribution, the same methods were run on n=52 real third-party meshes (Objaverse LVIS subset and ModelNet40), mapped to the six families through a class-to-family mapping frozen in writing before any evaluation and passed through visual quality control (full table, per-case scatter, and a qualitative gallery of the two representative outcomes in RP). Mean fit time remains millisecond-scale (9.8 ms per case, CPU). This is a sanity check, not a benchmark: the sample is small, the class mapping is approximate, and the evaluation reuses the synthetic-grid protocol on real geometry.

Three readings follow. First, the preregistered margin does not replicate on external meshes at this sample size: SPPA-MVFit scores 0.413 (CI $[0.359, 0.467]$) against Generic-MVFit 0.370 (CI $[0.335, 0.405]$), a paired +0.043 whose interval $[-0.007, +0.094]$ includes zero. Second, the nonsemantic visual hull dominates raw occupancy with clean real silhouettes (0.656, CI $[0.606, 0.704]$), and the trivial geometric primitives tell the same story: capsule (0.492), ellipsoid (0.485), and even the axis-aligned box (0.411) match or exceed SPPA-MVFit (0.413) here—outside the design distribution the family graph buys no volumetric advantage over shape-free baselines, and the value of SPPA is the semantic contract: role labels, bounded updates, and a telemetry-scale payload. Third, the failure modes are informative and consistent with Section 4.4: template–instance mismatch (horizontal water towers fitted by a vertical tower graph), open geometry (bicycle wheel spokes), and one LVIS mislabel retained as an out-of-distribution case (a giraffe labeled `horse`). When the prior does not match, it hurts; the contract's role is to detect that situation and degrade safely. Under mild morphology corruption the pattern persists (SPPA 0.380, Generic 0.357).

4.8 External neural generators: an input-modality mismatch demonstration

A secondary analysis, specified in writing before any measurement, compares SPPA-MVFit with four measured external neural image-to-3D generators, TripoSR [37], Hunyuan3D-2mini-turbo and the flagship Hunyuan3D-2 full model [45], and TripoSG [19], on sixty of the held-out cases (Table 7 and Figure 10; protocol in RP). Each generator received two inputs: a clean-crop shaded oblique render, its best condition, and the raw 96×96 top telemetry mask, the closest visual analog of the channel SPPA consumes. On these cases SPPA-MVFit reaches 0.561 mean voxel IoU at 9.2 ms CPU on this 60-case subset with a 1.45 kB update descriptor, against 0.128–0.231 for the measured neural outputs over seven of the eight method-by-input conditions—the eighth, TripoSG on the raw mask (0.002), is the input-parity artifact registered in RP and excluded from this range—whose meshes carry 28k–3.4M triangles and MB-scale payloads on an RTX 5090. Two observations sharpen the boundary. First, every generator with a parity-clean input comparison scores *higher* from the harsh top mask than from its generous clean-crop render (TripoSR and both Hunyuan3D-2 variants): single-view image-to-3D models hallucinate occluded structure, whereas a calibrated footprint constrains occupancy directly. In this asymmetric setting, telemetry-shaped evidence behaves not as a degraded image but as a different measurement channel. Second, the wave measures one operating point per system under deliberately asymmetric input channels, not an image-to-3D leaderboard; the flagship fifty-step Hunyuan3D-2 full model (0.148 oblique, 0.177

mask) sits inside the same band as its distilled mini/turbo variant, retiring the distilled-checkpoint caveat by measurement.

Table 7: Measured external neural wave (secondary, post-hoc; restored to the main text in the 2026-07-21 readability pass). Mean voxel IoU at 64^3 on the same sixty held-out cases (Hunyuan3D-2mini-turbo (a): n=58 after the two reported hard crashes), with parenthesized paired ΔIoU versus SPPA-MVFit. Triangles and payload are mesh-output means; generation time is warm per-case wall time on the same RTX 5090; VRAM is uncapped peak CUDA allocation. SPPA rows are CPU-only and their payload is the update descriptor, not a mesh. TripoSG (b) is the input-parity artifact of the wave protocol (registered in RP). Per-case values: `benchmarks/results/sppa_neural_flagship_wave.json`.

Method	Input	IoU	Triangles	Gen. time (ms)	VRAM (MB)	Payload (B)
SPPA-MVFit (ours)	top+side 96x96 telemetry masks	0.561	536	9.2	–	1,450
Generic-MVFit (context)	same masks	0.364 (-0.197)	1,280	11.4	–	1,433
Visual hull (context)	same masks	0.516 (-0.045)	51,432	0.2	–	32,768
TripoSR (a)	clean-crop shaded render	0.128 (-0.433)	28,145	384.1	1,869	1,468,998
TripoSR (b)	96x96 top mask	0.231 (-0.331)	46,053	379.4	1,870	2,454,199
Hunyuan3D-2mini-turbo (a)	clean-crop shaded render	0.157 (-0.403)	686,733	1350.9	4,591	10,513,018
Hunyuan3D-2mini-turbo (b)	96x96 top mask	0.171 (-0.390)	3,089,274	1915.5	4,747	46,494,183
TripoSG (a)	clean-crop shaded render	0.147 (-0.415)	697,881	9064.4	5,602	12,563,206
TripoSG (b)	96x96 top mask	0.002 (-0.559)	113,799	6281.8	5,409	2,049,198
Hunyuan3D-2 full (a)	clean-crop shaded render	0.148 (-0.413)	868,968	9264.3	5,650	13,037,869
Hunyuan3D-2 full (b)	96x96 top mask	0.177 (-0.385)	3,388,675	10005.6	5,936	50,848,377

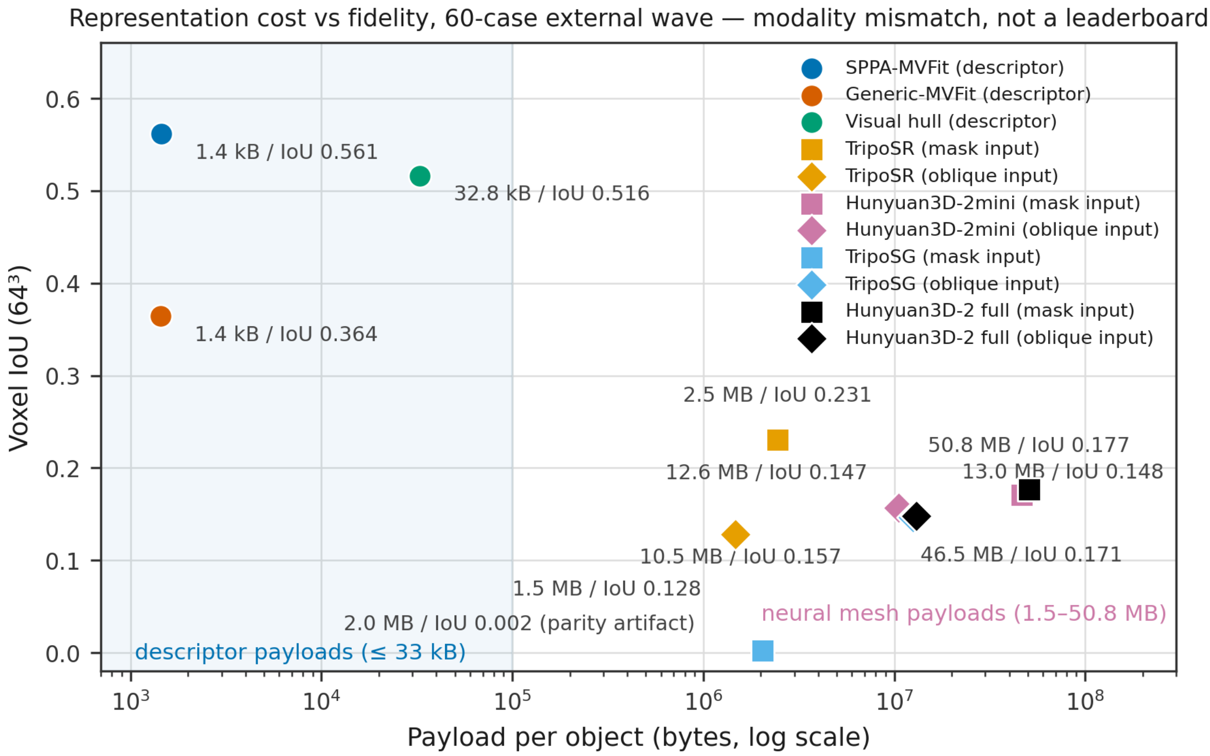

Figure 10: Operating-point comparison on sixty held-out cases: mean voxel IoU against per-object payload in bytes (log scale) for SPPA-MVFit, Generic-MVFit, visual hull, and the four measured neural generators under both input conditions. SPPA rows are CPU-only and their payload is the update descriptor, not a mesh; TripoSG's mask-input point (0.002) is the input-parity artifact registered in RP. The juxtaposition is an input-modality mismatch shown for operating-point context, not a leaderboard.

4.9 Real-image probes and deployment runtime

Real-image probes exercise the dual-input contract of Section 3.2: a YOLOE-26s open-vocabulary detector supplies approximate semantic evidence on the image path, while the tag path uses reviewed text only. Detector evidence is not treated as ground truth: weak labels are refined only to conservative families (the tractor-trailer worked example of Section 3.2), implausible footprint aspect ratios are clamped by family priors, and unsafe image-derived pose is withheld from the runtime descriptor. Figure 11 runs the full chain end to end on two real flight photos recorded for this paper (2026-07-21). The tower photo receives `electric pylon` (confidence 0.49, 38-point native mask) and enters the `power_tower` family: the geo-projected mask footprint (39.9×17.5 m, cables included) is implausible for a tower, so the gate clamps it through the vertical-structure aspect prior to $5.60 \times 5.60 \times 28.00$ m and the builder emits a 396-triangle lattice proxy. The tractor photo is mislabeled by the detector as `two-wheeled vehicle` (0.48), so it is routed to the conservative `generic_vehicle` family rather than a committed wrong token; the 7.1×6.0 m footprint is fused to $4.75 \times 2.50 \times 2.60$ m and a 576-triangle proxy. On the same crops and workstation, the measured neural generators degrade on these aerial views: TripoSR (6 GB cap) returns 26,836- and 39,700-triangle meshes at 0.49/0.09 s inference and 1.9 GB peak VRAM, but as amorphous blobs without tower or tractor structure, and Hunyuan3D-2mini-turbo fails with no surface on both crops (5 and 20 inference steps), while a ground-level control photo works (Figure 11d; per-case values in RP). Telemetry for these replays is declared assumed flight (45/35 m AGL, nadir camera), not measured; no 3D ground truth exists for the photos, so this fidelity reading is qualitative, not an IoU ranking. The four earlier archived probes (biker, mountain tower, mountain tractor, tractor-trailer), the probe grid, per-probe evidence, replay audits, and input-mode ablations are retained in RP.

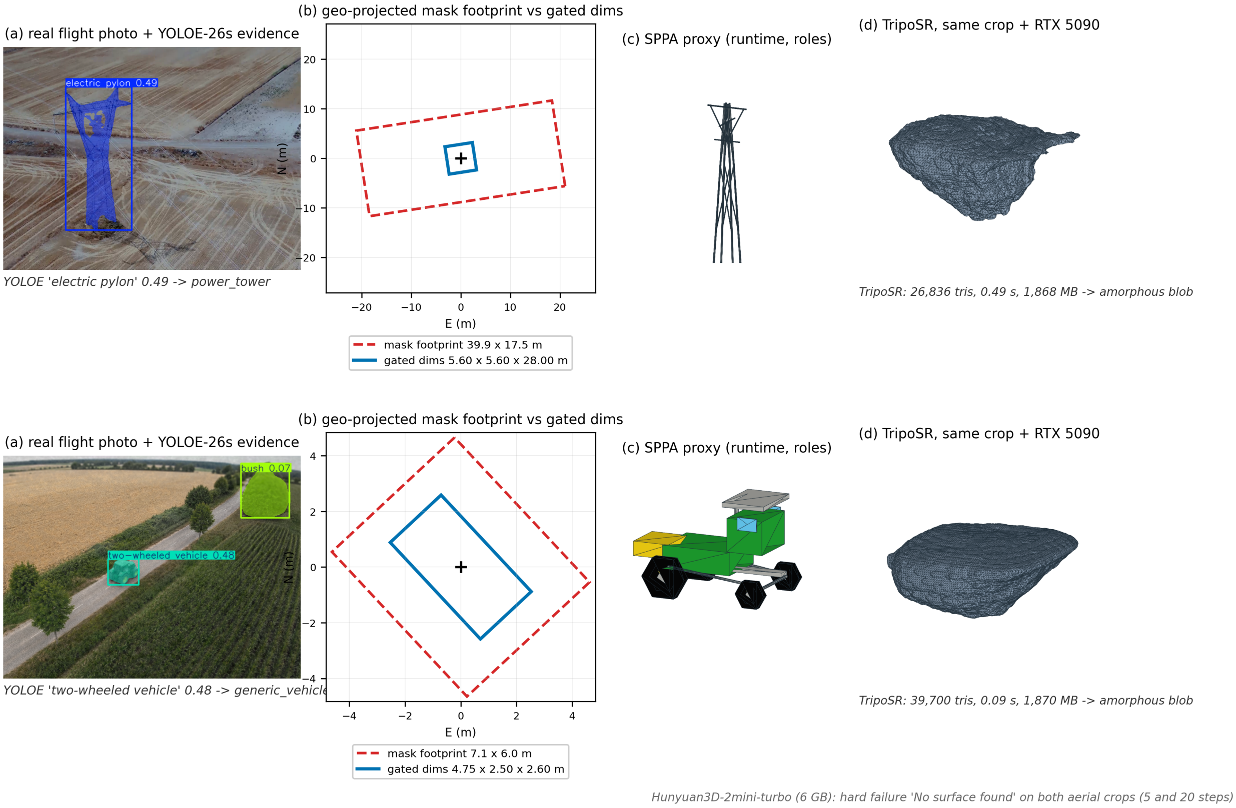

Figure 11: Real flight-photo probes end to end (exploratory, no 3D ground truth; declared assumed-flight telemetry). Rows: tower and tractor photos recorded for this paper. (a) YOLOE-26s evidence (bbox, native mask, confidence). (b) Geo-projected mask footprint (dashed red) against the gated metric dims (blue): the tower's 39.9 × 17.5 m footprint is clamped to 5.60 × 5.60 × 28.00 m by the family aspect prior; the tractor's 7.1 × 6.0 m is fused to 4.75 × 2.50 × 2.60 m. (c) The compiled runtime proxies (396 and 576 triangles): the detector mislabels the tractor as `two-wheeled vehicle`, which exercises the conservative `generic_vehicle` routing instead of a committed wrong token. (d) TripoSR on the same object crops (6 GB cap, RTX 5090): amorphous blobs (26,836 and 39,700 triangles); Hunyuan3D-2mini-turbo returns no surface on either aerial crop. Neural rows are qualitative context, not a ranking; measured IoU comparisons are the sealed synthetic wave of Section 4.8.

A verifier-gated open-label probe of 20 awkward labels verified 8/20 recipes with zero critical mismatches, and a role-preservation diagnostic recipes with zero critical mismatches, and a role-preservation diagnostic keeps cab and wheel priors fixed while cargo length changes. No 3D ground truth exists for these images, so no IoU is reported for them.

The deployment layer is the descriptor/update contract consumed by the optional Unreal Engine 5.7 backend. Table 8 condenses the runtime evidence, including the packaged scaling behavior (scaling figure in RP); the full Unreal/HISM matrix with run identifiers is in RP. Packaged replay at 100 synthetic objects passes a 33.3 ms desktop frame budget but misses 60 FPS; dense shape updates remain the limiting factor even under HISM batching, and curated assets remain preferable when a matched asset exists (shape-stream P95 6.789 ms vs. 22.445 ms on the two-label cow/tower subset).

Table 8: Condensed deployment runtime evidence (run identifiers omitted here; full matrix in RP). Python contract timings are not Unreal frame times, and packaged runs are short synthetic replays, not live flight telemetry. The link-budget row is byte accounting on measured replay traffic against a modeled video reference, not a measured radio link.

Evidence	Summary value	Boundary
Median SPPA-MVFit fit (sealed synthetic benchmark)	9.4 ms CPU per object	single-call wall clock, not frame time
Real-stream fit latency (Section 4.10)	11.8 ms median per detection	recorded 2–15 Hz stream
Python contract build (descriptor, update packet, scheduler decision)	P95 $\leq$ 0.52 ms	Python contract cost, not frame time
Measured SPPA JSON update traffic (no-op packets suppressed)	25.8–37.4 kB/s; modeled compressed video $\geq$250 kB/s (2–10 Mbps)	byte accounting on replay; video rate modeled, link not validated
Editor-Cmd and packaged recorded replay	27/27 payload applications accepted	recorded streams, no live network
Packaged 100-object replay	pose/shape frame P95 17.814/19.256 ms	desktop feasibility; misses 60 FPS P95
Packaged HISM dense-update scaling, 100/250/500 objects	shape frame P95 37.126/94.018/186.113 ms; pose frame P95 18.921/41.609/78.933 ms	dense updates exceed the 33.3 ms budget (shape from 100, pose from 250 objects); 60 FPS not reached
Packaged HISM partial update, 500 objects	pose/shape frame P95 16.528/29.787 ms	10% changed-track synthetic schedule

4.10 A real detector stream case study

The motivating scenario is a real detector running on a live georeferenced twin, and the value of this case study is precisely that its signal is degraded. As a post-hoc exploratory case study, a recorded 1,394-frame flight stream over the Ejea twin was re-analyzed end to end: a custom YOLO detector with real confidence noise and real confusions (cows hallucinated on tower textures), MAVLink telemetry, and exact simulator ground truth for the static tower anchors. Detections with locked telemetry yield 1,902 cases (308 tower, 848 cow, 746 biker detections). Every method receives the identical observation—an oriented ground footprint geo-projected from the detection box plus a monocular height estimate—so the comparison carries no input advantage, and wrong detector tokens are kept as a natural condition. Figure 12 maps the flight and its detections against the simulator ground-truth anchors.

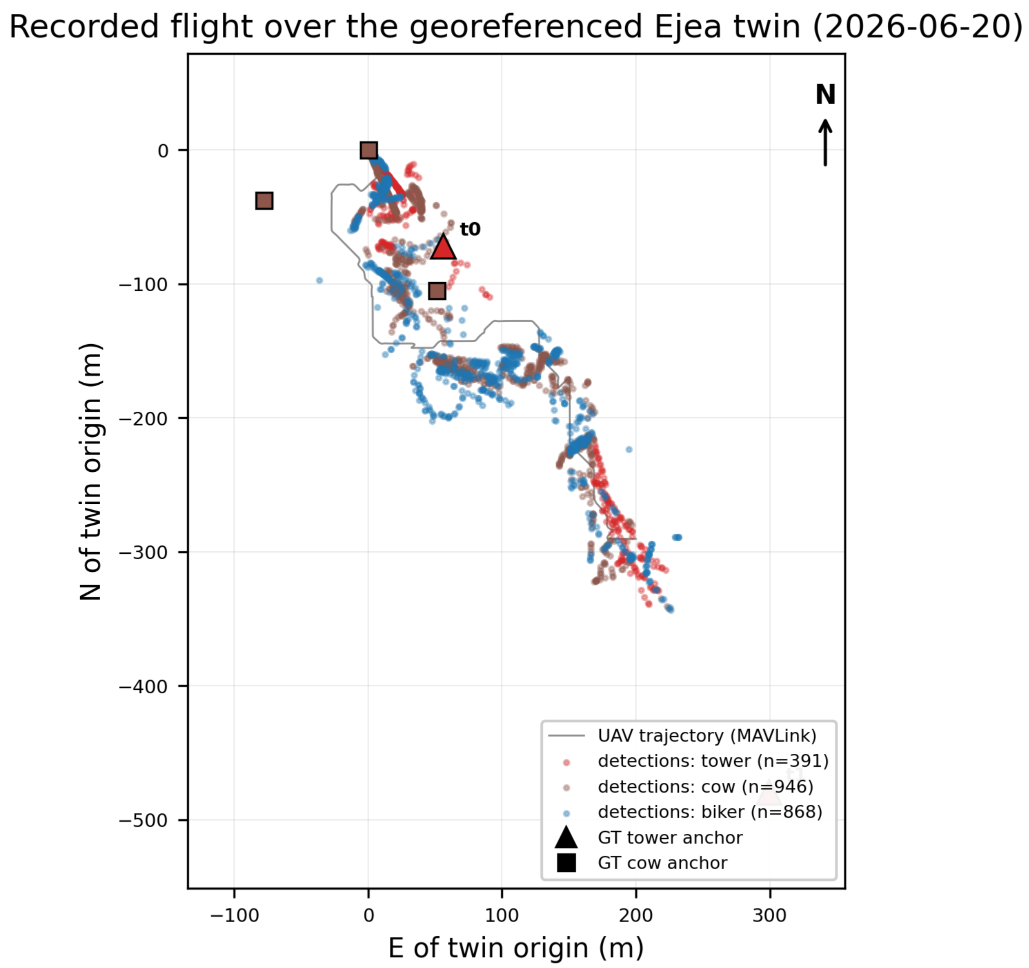

Figure 12: The recorded flight in georeferenced coordinates (twin-local ENU, meters). The UAV trajectory (MAVLink telemetry) and the raw detector outputs with geo-reference are plotted against the exact simulator anchors; only towers `t0`–`t1` fall inside the observed corridor, so the GT-matched analyses (RP) concentrate on them. The plotted detections are the raw stream events; the 1,902 cases of the case study are the telemetry-locked subset. Cow and biker detections are real detector outputs—many are hallucinations on terrain texture, kept deliberately as the natural wrong-token condition.

Three findings characterize the degraded-signal operating point (Table 9; Figure 13). First, the family prior buys 3D part structure at the cost of 2D image overlap on real evidence: SPPA-MVFit does *not* win the reprojection metric on this stream (0.298 median versus 0.33–0.45 across the nonsemantic baselines), because the monocular height estimate conflicts with the prior bounds (a 13.75 m tower column versus a ≈5 m observed extent). The stream carries no 3D ground truth, so the quantity the sealed benchmark measures—3D occupancy—cannot be scored here; what the stream adds is precisely the evidence-versus-prior tension that a ground-truth benchmark cannot see. Second, the tension is quantifiable and diagnostic: on the 138 GT-matched cases where the real detector emitted a wrong family token, refitting with the correct token collapses the 2D fit from 0.381 to 0.025—the correct prior cannot explain the misleading evidence that triggered the wrong label (token-arm table in RP). Third, the operating point is computationally viable: fit latency (11.8 ms median) stays far below the 2–15 Hz stream rate, even in this degraded mode. Localization tells a separate story and is outside this paper's scope: all methods share the same ≈33 m median 3D error, set by the monocular observation geometry of the low-altitude stream rather than by the fitter; the full localization analysis (localization and footprint columns, plan-view and error-identity panels) is retained in RP (`reproducibility/sppa_mvfit/`).

Token validation (post-hoc). The same stream provides a measured token-validation signal (protocol frozen in writing before any outcome was computed; routing arm and ROC in RP): on the 217 GT-matched cases, −confidence separates the 138 wrong-token from the 79 correct-token cases with AUC 0.847 (95% CI $[0.792, 0.898]$; bootstrap 10,000 resamples, seed 77157), and prior mismatch separates them perfectly, in the direction opposite to intuition: the cow/biker hallucinations receive small monocular heights ($\approx$1–2 m) that fit their wrong family priors suspiciously well, while true towers suffer severe monocular height underestimation ($\approx$5 m observed against a 25 m prior). The perfect separation is a no-observed-overlap result on these 217 cases, not a population claim; a suspiciously good prior fit is a measured wrong-token signature on this stream.

We read this study as the field counterpart of the robustness arm of Section 4.2: under a real degraded signal—real confidence noise, real wrong tokens, monocular height underestimation—the reconstruction path stays within its latency budget, and detector-token confidence and prior fit are measured validation signals the contract can act on when deciding whether to accept a family token, refine it to a conservative family, or fall back safely. The top-only arm of Section 4.6 quantifies the nadir regime this stream represents.

Table 9: Real detector stream case study (exploratory post-hoc, not sealed): 1,902 detections from a 1,394-frame flight over the georeferenced twin with a real custom detector, real telemetry, and exact tower-anchor ground truth. Reprojection and latency columns cover all cases. The localization and footprint columns (GT-matched subset, n=217, wrong detector tokens kept as a natural condition) are retained with the full table in RP: all methods sit at $\approx$33 m median 3D error, set by the monocular observation geometry. SPPA-MVFit trades 2D image overlap for 3D part structure (the stream has no 3D ground truth).

Method	2D reproj. IoU med. [P25, P75]	Latency (ms)
SPPA-MVFit (this work)	0.298 [0.218, 0.392]	11.8
Generic-MVFit	0.423 [0.394, 0.448]	13.4
OBB	0.446 [0.398, 0.492]	0.22
AABB	0.330 [0.299, 0.357]	0.25
Visual hull (non-semantic)	0.446 [0.398, 0.493]	0.50
Capsule	0.445 [0.422, 0.466]	1.31

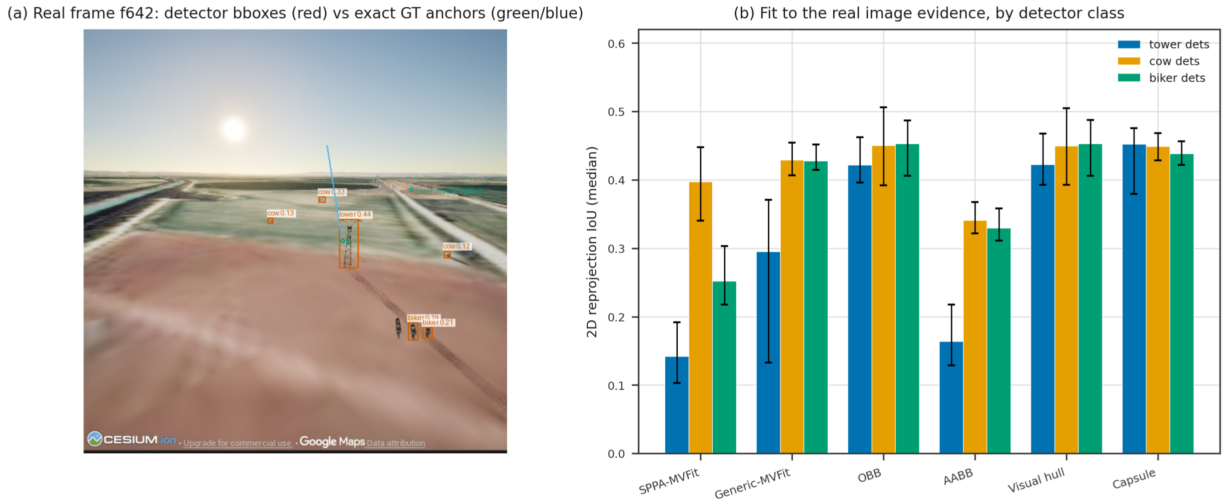

Figure 13: Real stream case study, degraded-signal view. (a) Frame 642 with detector boxes and exact ground-truth anchors reprojected through the same camera model. (b) 2D reprojection IoU against the real image evidence, by detector class: SPPA-MVFit does not win this 2D metric; the family prior buys 3D part structure instead. The plan-view and observation-error panels that document the (out-of-scope) localization behavior are retained in RP.

4.11 Reconstruction fidelity across view angles in the twin (E11, exploratory)

The confirmatory study and the real-stream case study leave one mission question open: does the reconstruction stay faithful across the view angles a flight actually delivers, against exact 3D ground truth? E11 is an exploratory post-hoc study (not confirmatory; its protocol was frozen before any outcome) on the georeferenced twin. It captures 308 twin frames (640×640, FOV 70°) of the 11 real lattice towers of the Ejea map along three view rings per tower—oblique 30° and oblique 45° (12 azimuths each) and nadir (4 azimuths, +60 m)—runs the real YOLO detector of Section 4.10 on those frames, and fits all methods with positions locked to the exact simulator ground truth. Only reconstruction fidelity is evaluated; no localization metrics are computed or claimed. The evidence is hybrid: simulated twin imagery, a real detector, and exact simulator ground truth.

Table 10 and Figure 14 summarize the study. Inference is by tower-cluster bootstrap (11 towers as clusters, 10,000 resamples, seed 20260720, post-hoc): detections within a tower are not treated as independent. On the oblique rings—the views a real pass delivers—SPPA-MVFit leads every baseline: mean voxel IoU 0.118 ($[0.109, 0.125]$) at 30° and 0.087 ($[0.061, 0.120]$) at 45°, against 0.040 and 0.037 for the best baseline (capsule); the paired tower-clustered deltas over every baseline are all positive (+0.050 to +0.094), with $P(\Delta \leq 0) = 0$ in all resamples. On the correct-token subset the SPPA fits reach 0.125 ($[0.122, 0.127]$, n=140) at 30° and 0.147 ($[0.145, 0.150]$, n=89) at 45°. The wrong-token rate rises from 6.0% (9/149) at 30° to 42.2% (65/154) at 45°; the apparent correct-token inversion ($0.147 > 0.125$) is therefore read with caution—at 45° nearly half of the detections are filtered out as wrong tokens, so the correct-token subset concentrates on the clearest views, and part of the inversion is this dilution effect rather than a pure view-angle gain. The monocular height estimate that anchors the fit averages 21.3 m across tower detections against an exact ground-truth height of 20.71 m. At nadir the tower is invisible to the detector (zero tower detections on that ring): the nadir rows fit the spurious non-tower tokens that do arrive, and there the wrong family token hurts exactly as the wrong-family analysis of Section 4.4 predicts—shape-free boxes score higher on those cases. The nadir ring therefore quantifies the footprint-only degraded regime rather than a reconstruction regime.

Two caveats temper the oblique result. First, the absolute IoU values are modest: a lattice tower is mostly air at 64^3 (mean GT occupancy $\approx$3.2k of 262,144 voxels), so voxel-exact agreement is dominated by sub-voxel member placement. Second, cross-view consistency cuts the other way superficially: boxes are trivially more self-consistent across views (pairwise proxy IoU up to 0.636–0.677 for AABB) because a box fitted from any view is nearly the same box—while their IoU against ground truth stays at 0.02–0.04. SPPA proxies agree at 0.290 (30°) and 0.273 (45°) across views, and the per-tower SPPA consensus proxy reaches 0.099 ($[0.095, 0.104]$) against ground truth: consistency is reported alongside fidelity, not in its place.

Table 10: E11 oblique twin wave (exploratory post-hoc, not confirmatory; positions locked to ground truth; hybrid evidence: simulated imagery, real detector, exact simulator GT). Mean 3D voxel IoU against the exact tower meshes at 64^3 with tower-cluster bootstrap 95% CIs (11 towers as clusters, 10,000 resamples, seed 20260720, post-hoc; detections within a tower are not treated as independent). Capsule is the best token-free baseline on both rings. Wrong-token rows report Wilson 95% CIs with the tower-cluster CI in braces. The nadir block is summarized in prose; the cross-view blocks are in RP (`reproducibility/sppa_mvfit/`).

Ring	Method	mean 3D IoU [95% CI] / median	n	SPPA−method [95% CI]
oblique30	sppa_mvfit	0.118 [0.109, 0.125] / 0.125	149	
	generic_mvfit	0.044 [0.042, 0.046] / 0.045	149	+0.074 [+0.066, +0.081]
	obb	0.031 [0.030, 0.033] / 0.031	149	+0.087 [+0.076, +0.095]
	aabb	0.023 [0.022, 0.025] / 0.019	149	+0.094 [+0.084, +0.103]
	visual_hull	0.031 [0.030, 0.033] / 0.031	149	+0.086 [+0.076, +0.095]
	capsule	0.040 [0.038, 0.042] / 0.040	149	+0.078 [+0.068, +0.086]
oblique45	sppa_mvfit	0.087 [0.061, 0.120] / 0.127	154	
	generic_mvfit	0.024 [0.022, 0.026] / 0.026	154	+0.063 [+0.039, +0.094]
	obb	0.035 [0.025, 0.044] / 0.022	154	+0.052 [+0.017, +0.095]
	aabb	0.033 [0.023, 0.040] / 0.021	154	+0.054 [+0.021, +0.097]
	visual_hull	0.035 [0.025, 0.044] / 0.022	154	+0.052 [+0.017, +0.095]
	capsule	0.037 [0.028, 0.044] / 0.028	154	+0.050 [+0.017, +0.091]
Correct-token subset (cluster-bootstrap CIs, tower-level resampling)				
oblique30 (correct token)	sppa_mvfit	0.125 [0.122, 0.127] / 0.126	140	
	generic_mvfit	0.045 [0.044, 0.047] / 0.046	140	+0.080 [+0.076, +0.083]
oblique45 (correct token)	sppa_mvfit	0.147 [0.145, 0.150] / 0.148	89	
	generic_mvfit	0.029 [0.028, 0.030] / 0.029	89	+0.118 [+0.116, +0.121]
Wrong-token rate per ring (Wilson 95% CI; cluster CI in braces)				
oblique30	wrong token	9/149 = 0.060 [0.032, 0.111] {[0.007, 0.129]}	149	
oblique45	wrong token	65/154 = 0.422 [0.347, 0.501] {[0.194, 0.602]}	154	

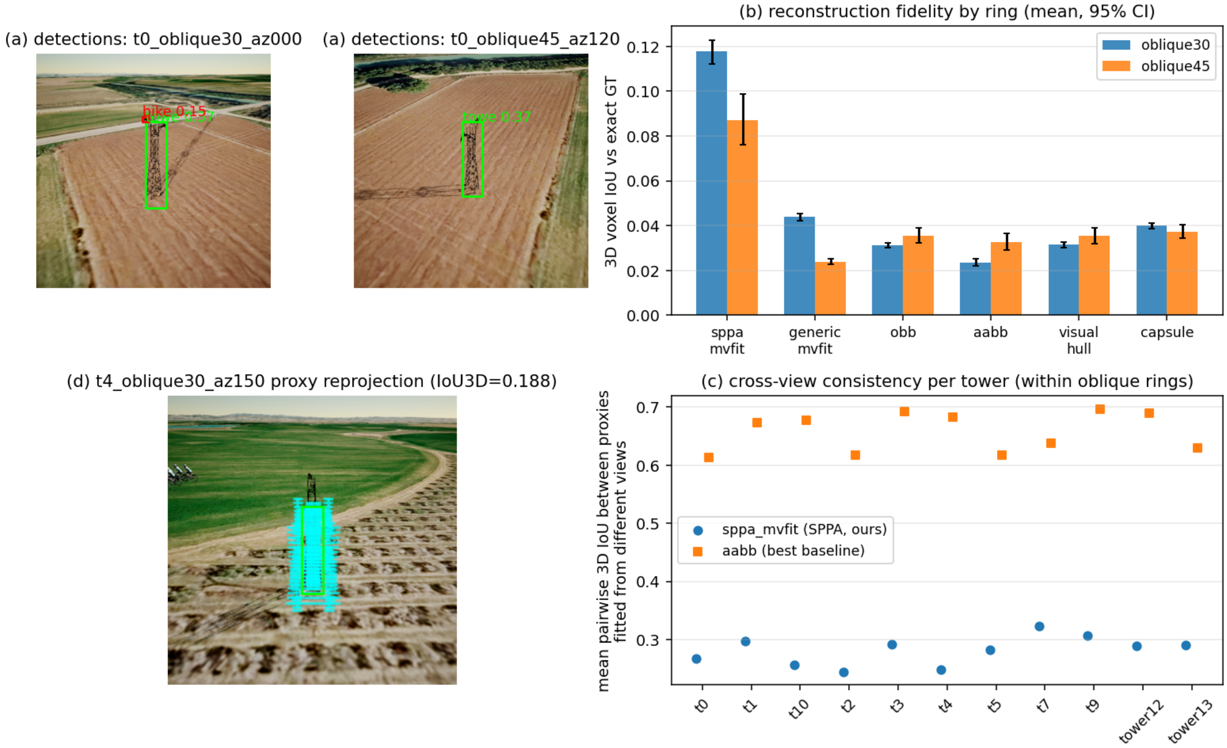

Figure 14: E11 oblique twin wave (exploratory post-hoc; positions locked to ground truth). (a) Sample real-detector detections on simulated twin frames. (b) 3D voxel IoU against exact simulator ground truth by view ring: SPPA-MVFit leads on the oblique rings; the nadir ring carries no tower detections and quantifies the degraded regime. (c) Cross-view consistency per tower: boxes are trivially more self-consistent while remaining far from ground truth. (d) An SPPA proxy reprojected onto the source view.

4.12 Camera-less sensing in the twin: a simulated-LiDAR demonstration (E14, exploratory)

E14 asks whether the same reconstruction contract can be fed when the camera is useless (night, fog, smoke): an exploratory post-hoc demonstration whose returns are SIMULATED LiDAR-class line raycasts inside Unreal Engine 5.7 PIE on the Ejea twin—not hardware, and with no camera anywhere in the perception path. Positions are locked to ground truth; the scope is reconstruction only. Five poses per tower (55 scans) fire a 361×16 azimuth–elevation ray fan (5,776 rays per scan); degradation is applied offline with frozen seeds so a clean arm (5% dropout, 2 cm range noise, 150 m max range) and a degraded heavy-fog arm (50% dropout, $\times4/\times2$ angular downsampling to 728 rays per scan, 5 cm noise, 100 m max range) are exactly paired. A GT-free vertical-structure clustering stage detects each tower and emits only an oriented footprint, yaw, and height; the sealed SPPA-MVFit production mode and the nonsemantic baselines of the real-stream study (Section 4.10) consume exactly that.

Table 11 and Figure 15 give the outcome, and the reading is threefold. First, the camera-less contract works at the envelope level: the recovered footprint is 7.9×4.0 m against a ground-truth 7.42×3.43 m across all 11 towers, yaw is within $\approx6°$, height within $\approx5\%$ (median), and footprint IoU is 0.43. Second, voxel-exact 3D IoU is ≈0.08 for every method, SPPA included: the lattice occupies only $\approx$1,100 of 262,144 voxels, so even the oriented box's structural ceiling is $|GT|/|OBB| \approx 0.08$, and the metric at 64^3 is dominated by sub-voxel member placement. SPPA-MVFit does not beat the baselines on this truth (clean arm: SPPA 0.081 $[0.076, 0.085]$, generic 0.084, capsule 0.083); with a 1-voxel tolerance (post-hoc secondary measure) SPPA reaches 0.168 against capsule's 0.190. What SPPA buys here is economy, not IoU leadership: $\approx$2.8k occupied voxels against the box's 14.3k at comparable IoU (voxel precision 10.5% versus 6.7%)—its mass lands on structure while the box's mass is mostly air. Third, degradation is measured rather than assumed: at 50% dropout with angular downsampling, 4 of 11 towers (`t1`, `t2`, `t10`, `tower13`, for all six methods) are lost to detection failure—that detection loss is the real cost of the degraded arm. The seven survivors are pairwise comparable to their clean-arm fits (paired delta $+0.002$ $[-0.002, +0.006]$, n=7); degraded-arm means are computed on survivors and suffer survivor bias, so the paired comparison is the one reported. Inference is by tower-cluster bootstrap (11 towers as clusters, 10,000 resamples, seed 77157); the cluster intervals barely differ from the original ones, whose bootstrap was already effectively per tower.

The sensor-model limits: UE simple-collision hulls stand in for laser returns, and beam divergence, multi-return behavior, reflectivity, and motion distortion are not modelled. E14 therefore supports the sensor-agnostic contract of Section 3.2 as a simulated demonstration only; it is not evidence of hardware LiDAR performance.

Table 11: E14 LiDAR twin wave (exploratory post-hoc; SIMULATED LiDAR-class returns from UE raycasts, no camera in the path; positions locked to ground truth). Mean 3D voxel IoU against the welded tower mesh at 64^3, clean versus degraded (heavy-fog) arms, with tower-cluster bootstrap 95% CIs (resampling unit: tower; clean n=11 towers with 0 detection failures; degraded n=7 fitted towers after 4/11 detection failures—`t1`, `t2`, `t10`, `tower13`, failed for all six methods; 10,000 resamples with replacement, seed 77157). The degraded-arm means are computed on the 7 survivors and carry survivor bias; the paired clean-versus-degraded delta for SPPA-MVFit is $+0.002$ $[-0.002, +0.006]$, n=7.

Method	clean mean 3D IoU [95% CI]	degraded mean 3D IoU [95% CI]
sppa_mvfit	0.081 [0.076, 0.085]	0.082 [0.076, 0.088]
generic_mvfit	0.084 [0.082, 0.087]	0.096 [0.094, 0.099]
obb	0.066 [0.064, 0.068]	0.087 [0.074, 0.108]
aabb	0.057 [0.053, 0.061]	0.059 [0.052, 0.068]
visual_hull	0.065 [0.063, 0.068]	0.088 [0.076, 0.108]
capsule	0.083 [0.082, 0.085]	0.105 [0.092, 0.128]

Resampling unit: tower. clean: n=11 towers, 0 detection failures; degraded: n=7 fitted towers after 4/11 detection failures (t1, t2, t10, tower13, failed for all methods). 10,000 cluster bootstrap resamples over towers with replacement, seed 77157.

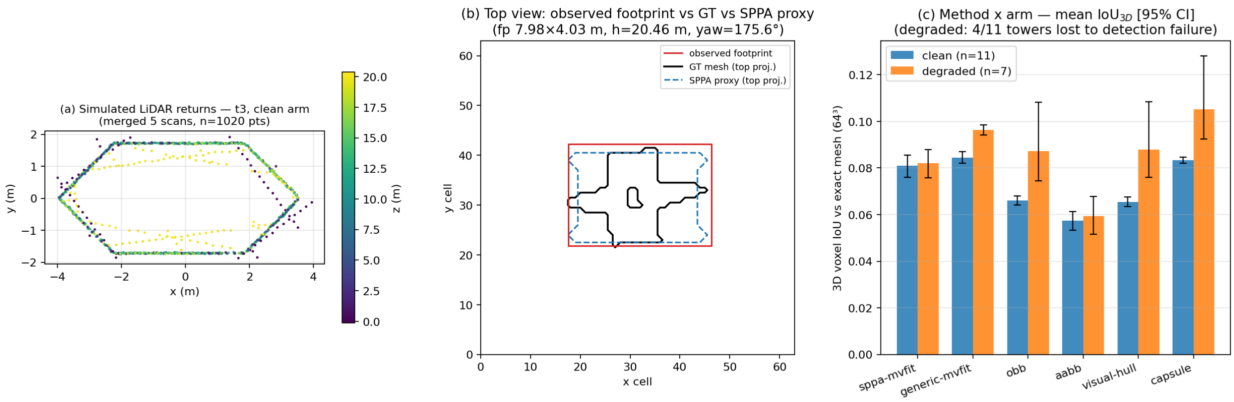

Figure 15: E14 simulated-LiDAR demonstration (exploratory post-hoc; returns are UE raycasts, not hardware). (a) Simulated LiDAR-class returns on tower t3. (b) Recovered footprint and SPPA proxy against the ground-truth top silhouette: the camera-less contract recovers the envelope (footprint, yaw, height) reliably. (c) Voxel-exact 3D IoU by arm and method: all methods sit at the lattice's sub-voxel structural ceiling (≈0.08); SPPA's advantage in this setting is proxy economy, not IoU leadership.

5 Discussion

The motivating mission is the stale-twin delta: when the image link cannot show the scene, the twin must still absorb the detections that arrive and reconstruct the objects its static layers lack, per object and at telemetry bandwidth. The confirmatory result answers the preregistered question behind that mission: under identical calibrated silhouettes and optimizer budget, a frozen family part graph improves lightweight occupancy over a generic eight-part graph by a large and stable margin. The decomposition of Section 4.3 shows that the knowledge representation, not a clever optimizer, is the active ingredient; secondary gains over text-only SPPA show that silhouette evidence helps when the same family graph is held fixed. The robustness arm is what makes the result mission-relevant: the margin holds under every preregistered corruption of the observation, mask noise costs the advantage only $\approx$0.001 IoU, and even moderate morphology corruption leaves 0.118, far above the prespecified margin—the path tolerates a weak or noisy signal, and only evidence that contradicts the prior measurably degrades it. This degraded-sensing evidence is asymmetric by construction: it combines a synthetic-corruption robustness arm, a simulated-twin demonstration, and one recorded stream case; no hardware measurement of degraded operation exists.

The contract is what remains differentiating at the measured frontier, and the boundary analyses are as informative as the headline. A wrong family token is worse than no prior at all, which turns fallback routing from an engineering convenience into a measured requirement. The adversarial stratum shows the advantage can be destroyed rather than being structurally guaranteed: violating the structural prior while preserving the semantic class destroys a third of the effect and produces identifiable tie zones where no axis-aligned part graph can win. The side view carries nearly all of the 3D information, yet it is unavailable from a pure nadir pass; the top-only arm quantifies exactly that nadir regime and still beats the generic dual-view fit. On external real meshes the margin does not replicate at this sample size, and shape-free baselines match or exceed SPPA on raw occupancy; what remains differentiating outside the design distribution is the contract—role labels, bounded updates, and a telemetry-scale payload—not volumetric leadership. Visual hull can approach SPPA-MVFit IoU because it is free to materialize denser occupancy without an eight-part role-labeled actor contract; the part-query task quantifies what that contract buys, since the hull covers but cannot select (query F1 0.434 versus 0.111). Finally, the real detector stream exercises the degraded-signal regime end to end: the fit stays far inside the stream's latency budget, the family prior trades image overlap for 3D structure under real evidence, and token confidence and prior fit are measured validation signals for accepting, refining, or rejecting a family token.

Five threats to validity remain binding.

- *Design and estimand.* The primary comparison is structurally favored by design—the family graphs encode the same part topology that the CSG-ID generators instantiate—so preregistration quantifies the effect but cannot make it surprising. The implicit-OOD stratum is the in-benchmark counter-evidence: a generator that does not share that topology keeps a 0.172 margin; the rest of the load-bearing evidence that the advantage is real, destructible, and bounded is the boundary map (wrong-family arm, adversarial stratum, external meshes, real stream), not H1 alone. The same team authored the family graphs and the generic competitor; the sensitivity variants of Section 4.6 bound, but do not eliminate, the resulting baseline-design risk.

- *Data.* The held-out set is synthetic and developer-held-out; positive OOD-stratum differences do not license claims about real UAV masks, detectors, or metric flight geometry. Family tokens are supplied, so the experiment is family-conditioned rather than open-set, and silhouettes are clean orthographic projections plus prespecified corruptions, not real detector masks.

- *Sensors.* The camera is the validated sensor: LiDAR interchangeability and night, fog, or smoke operation are evaluated only through the prespecified synthetic mask and morphology corruptions and the simulated-twin arms, not measured on hardware or in degraded weather.

- *Twin studies.* E11 and E14 are exploratory post-hoc studies, not confirmatory tests: simulated twin imagery (hybrid evidence with a real detector and exact simulator ground truth in E11) and simulated LiDAR-class raycast returns (E14), positions locked to ground truth in both. The E14 sensor-model simplifications and the degraded-arm survivor bias are given in Section 4.12; for thin lattices, voxel-exact IoU at 64^3 sits at a sub-voxel structural ceiling (≈ 0.08) for every method, so small IoU differences there are not reconstruction-quality rankings.
- *Runtime and stream.* "Instant" is per object (9.4 ms median fit); dense packaged shape updates do not reach 60 FPS and remain the declared limiting factor. Timing is local CPU wall-clock, not flight-laptop Unreal frame time; packaged replays are short synthetic streams. The real-stream study is one recorded flight with one custom detector, monocular low-altitude views, no 3D ground truth, and 2D reprojection metrics only; its measured operational signal is token validation (AUC 0.847 $[0.792, 0.898]$ for $-$confidence; 1.000 for $-$prior-mismatch on 217 matched cases of one stream—no observed overlap, not a population claim).

Within these boundaries the supported claim is narrow: on the held-out synthetic test, family conditioning improves occupancy beyond the prespecified margin; the shared generator supports text-only and silhouette-fitted modes; and SPPA descriptors can be consumed by an optional Unreal backend that coexists with curated assets. Real-UAV reconstruction accuracy, universal open-set geometry, operator benefit, and neural image-to-3D ranking remain outside the claim.

Three directions are left to future work. First, the frozen coordinate-descent fitter could be replaced by population-based or differentiable-rendering fitting, and the iterative fit could be amortized by a small learned mask-to-θ predictor, keeping the representation and contract while moving per-case cost toward microseconds; the primitive vocabulary could likewise be extended toward superquadrics. Second, the external gap measured here calls for public real-data tests with metric ground truth and detector-derived silhouettes rather than assumed calibration. Third, any operator-facing benefit requires a dedicated human-factors study; the present paper contains none.

6 Conclusion

SPPA-MVFit targets the moment a UAV digital twin must be updated under a weak signal: the static Cesium layer is stale, the image link does not suffice, and the detections that do arrive must become explicit per-object volumes at once. Family-conditioned SPPA-MVFit shows that a reviewed semantic part graph is a measurable occupancy prior for that instant per-object reconstruction: the preregistered equal-budget comparison passes its prespecified margin on the held-out synthetic test, and the margin is maintained under every preregistered observation corruption. Most of the advantage is the family-graph representation rather than the fitter; a wrong family token is worse than no family prior at all; the margin does not replicate on external meshes at this sample size; and adversarial instances that violate the structural prior destroy a third of the advantage, exposing identifiable tie zones. The durable contribution is the twin-delta update contract—family-conditioned roles, bounded in-place updates, a compact update descriptor at telemetry-scale update traffic, and a measured part-query value (query F1 0.434 versus 0.087–0.145 across occupancy and generic controls)—with declared limits: camera-validated, LiDAR and night/fog/smoke operation evaluated only through prespecified synthetic corruptions and simulated-twin demonstrations (E14 for camera-less sensing; E11 for cross-view fidelity), not measured on hardware, dense packaged shape updates below 60 FPS, and position assumed given. The next steps are external public data with metric ground truth, detector-derived silhouettes, and, only if claimed, operator studies.

Data Availability

The SPPA-MVFit reproducibility package, protocol amendments, NIST-bound seed manifest hashes, and confirmatory summaries are included under `reproducibility/sppa_mvfit/`. The preregistration, amendment log (Amendments 01–05), NIST-beacon seed manifest, SHA-256 freeze records, full per-family and secondary tables, and reproduction commands (`test_contract.py`; `verify_package.py --development`; `reproduce_sppa_mvfit_paper.py --strict` → 0 blockers) are included in the package. Large Unreal artifacts and historical engineering logs remain supporting material and are not required to recompute the primary H1 tables.

Code Availability

The fitting, evaluation, and table-export code used to produce the reported results is part of RP above and will be released in a public repository upon publication.

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